30th International Biology Olympiad
SZEGED, HUNGARY

Theoretical Exam 2

Afternoon

18th July 2019

COUNTRY
LANGUAGE

Theoretical Exam Afternoon

General Instructions

This exam consists of 38 questions and lasts 180 minutes.

- Cell Biology (Q1-7)
- Animal Anatomy and Physiology (Q8-16)
- Plant Anatomy and Physiology (Q17-22)
- Genetics and Evolution (Q23-30)
- Ecology (Q31-34)
- Biosystematics (Q35-36)
- Ethology (Q37-38)

1. **Please remember to attach your BARCODE sticker to all pieces of paper on the answer sheet.**
2. Write your answers in the separate answer sheet provided. **Only answers given in the answer sheet will be considered**.
3. Stop answering and put down your pencil immediately when the bell rings signalling the end of the exam.
4. No paper, materials or equipment should be taken out of the exam room.

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Cell Biology

Q1

The nematode *C. elegans* develops as either a male having only one sex chromosome (XO) or a hermaphrodite with two sex chromosomes (XX). When maintained in groups containing both sexes, hermaphrodites live significantly longer than males. Two signalling pathways, the IGF-1 signalling and the sex determination pathways play pivotal roles in the control of *C. elegans* ageing.

DAF-2 and DAF-16 are elements of the IGF-1 signalling pathway. Loss-of-function mutations in their genes were introduced, and the lifespan of the mutant animals of the two sexes were measured (Figure 1).

In all graphs, NS = not significant difference; *** = significant difference.

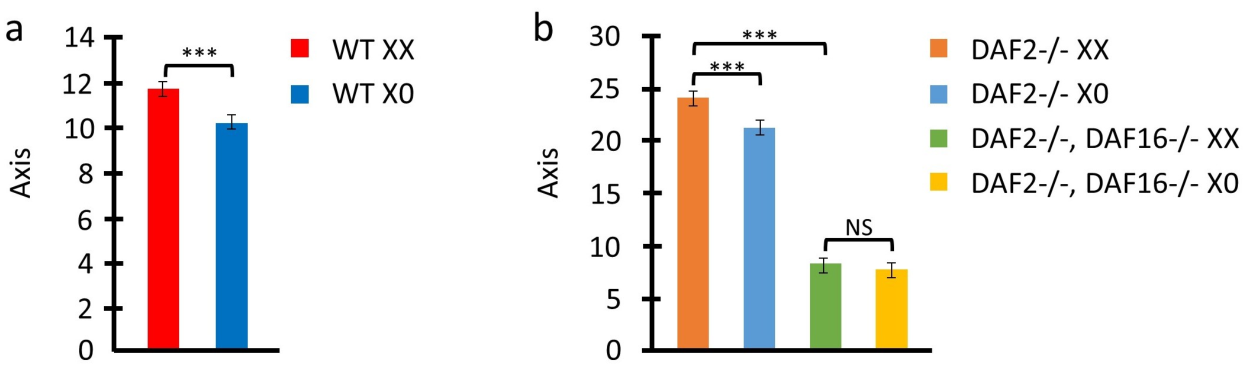


Fig.1. Axis = Mean lifespan (days). WT = Wild-type.

To test whether the nematode sex-determination cascade is involved in the sex-specific regulation of aging, researchers modified the activity of two elements of this pathway, TRA-1 and TRA-3. The mutations of transgenic hermaphrodites were as follows in Figure 2.

- MUT1 - TRA-1-/-
- MUT2 - TRA-1 low activity, with some function
- MUT3 - TRA-1 hyperactive function
- MUT4 - TRA-3 loss of function
- MUT5 - TRA-1 hyperactive function AND TRA-3 loss of function

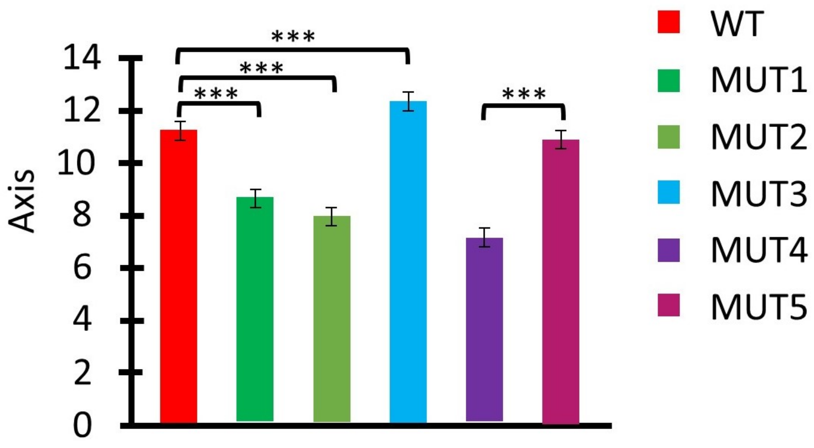


Fig.2. Axis = Mean lifespan (days). WT = Wild-type.

TRA-1 was identified as a transcription factor. The expression of DAF-16 was measured in various TRA-1 mutants to assess the functionality of a potential TRA-1-binding site in the DAF-16 gene (Figure 3).

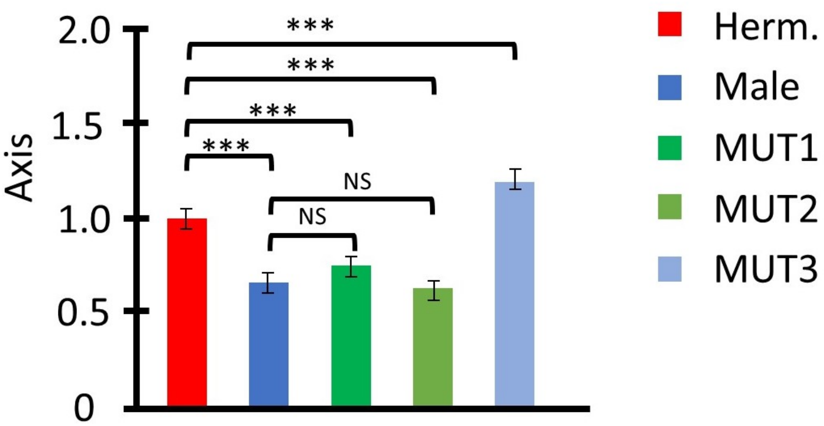


Fig.3. Axis = Relative DAF-16 expression level.

Q.1.1 Based on the information above, create a model of protein interactions that affect lifespan (LS). Use the options below (A-D) to fill in the boxes in the flowchart and indicate in the circles by the arrows whether the interaction is stimulatory (put **+**) or inhibitory (put **-**).
Options:
A. TRA-1
B. TRA-3
C. DAF-2
D. DAF-16

Q2

The ATP content of an average muscle fibre at rest is very low. Under short, high energy demand exercise (e.g. a 100 m sprint), the initially very low ATP levels in the muscle fibres are buffered by phosphocreatine (PCr, Figure 1).

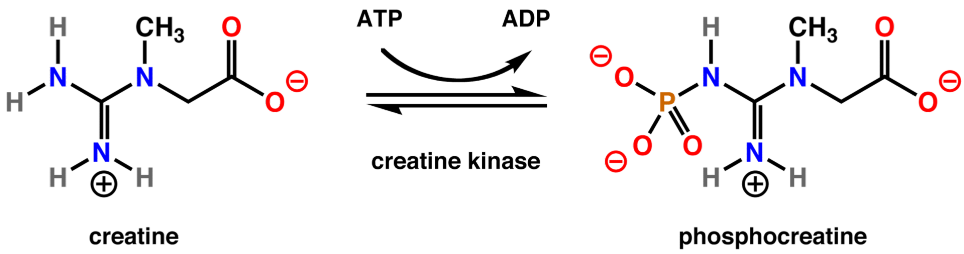


Fig.1

We can measure levels of phosphocreatine and ATP in a working tissue using 31P NMR spectroscopy, through the height of the corresponding NMR peak. Figure 2 shows the NMR-spectroscopy results of muscle intracellular fluid from a subject who had done 2 minutes of vigorous exercise. A: before exercise, B: first minute of exercise, C: last minute of exercise, D: after exercise; I, II and III are peaks that represent the three phosphate groups in ATP molecules, IV: PCr, V: Pi, VI: phospho-monoesters.

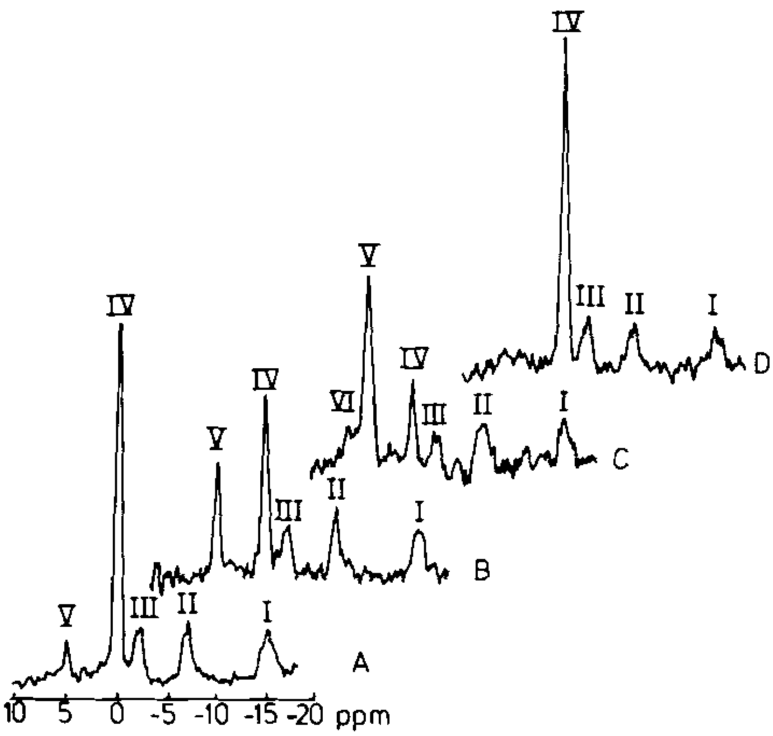


Fig.2

Immediately when the actin-myosin interaction starts producing ADP, the activity of the enzyme adenylate kinase, which catalyses the near-equilibrium reaction 2 ADP $\longleftrightarrow$ ATP + AMP becomes significant. Besides the generation of the negligible amount of available ATP, the reaction is essential because both ADP and AMP can allosterically upregulate glycolytic enzymes – and therefore prepare the muscle for longer exercise.

Based on the information above, indicate with an **X** if each of the following statements is true (T) or false (F).

Q.2.1 Hydrolysis of the high energy bond of PCr can be used directly as energy source of enzymes.

Q.2.2 The relatively constant ATP concentrations shown on the NMR results are due to the buffering effect of PCr.

Q.2.3 The phosphocreatine system is the main energy source in marathon runners.

Q.2.4 AMP and ADP levels are similarly sensitive indicators of energy status (as shown by ATP concentration).

Q3

Abscisic acid (ABA)-and salicylic acid (SA) are involved in stomatal closure. It's known that the Ca^{2+} -independent protein kinase OST1 and Ca^{2+}-dependent protein kinases (CPK3 and CPK6) are key for abscisic acid induced activation of an anion channel SLAC1 and stomatal closure. SLAC1 has two phosphorylation sites: S59 and S120 are both serines. Coexpression of the CPKs or OST1 together with SLAC1 genes in an observed cell leads to the following result:

Col 1				Col 2		Col 3
CPK3	CPK6	OST1	SLAC1	S59	S120	An
–	–	–	+	@	@	@
+	–	–	+	@	@	@
–	+	–	+	@	@	@
+	+	–	+	***	@	***
–	–	+	+	@	***	***

Table 1. Col 1 = Expression of . Col 2 = Phosphorylation of . Col 3 = Current. An = Anionic. '+' = expressed. '–' = not expressed. '@' = does not happen. '***' = happens

The effect of salicylic acid (SA) on stomata closure of wild type (WT) *Arabidopsis thaliana* was determined in a standardized way. Salicylic acid was added to a leaf fragments and stomata opening was measured. The stomatal assay was also carried out with some KO-mutant lines. The results are shown below. (*– significant change, ns – non-significant change)

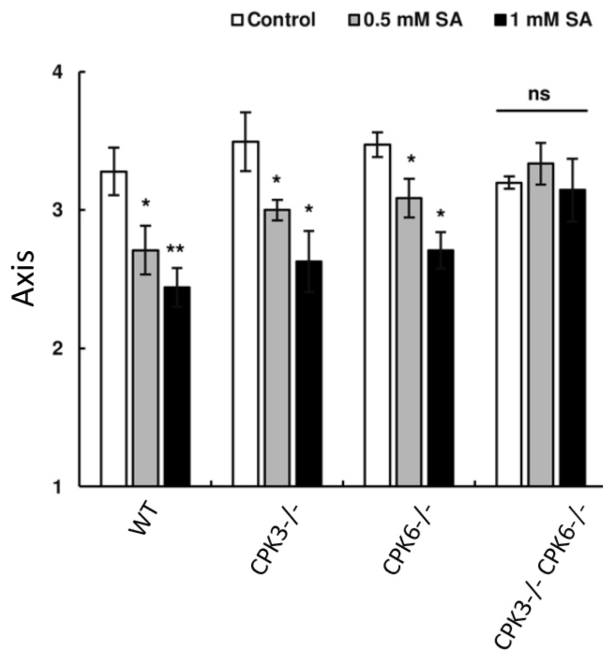


Fig.1. Axis = Stomatal aperture (μm). WT = Wild-type.

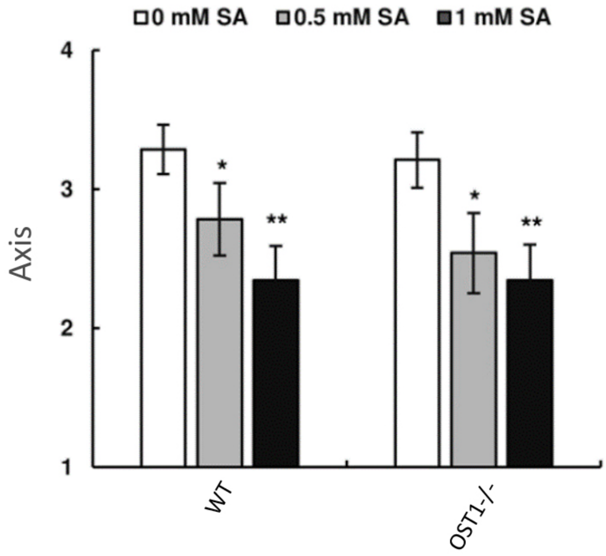


Fig.2. Axis = Stomatal aperture (μm). WT = Wild-type.

For the following questions indicate your answer (A-I) with an **X.**

A. CPK3

B. CPK6

C. OST1

D. either CPK3 or CPK6

E. either CPK3 or OST1

F. either CPK6 or OST1

G. CPK3 and CPK6

H. CPK3 and OST1

I. CPK6 and OST1

Q.3.1 Which of the kinase(s) (A-I) is/are required for the phosphorylation of site S59 in SLAC1?

Q.3.2 Which of the kinase(s) (A-I) is/are required for the phosphorylation of site S120 in SLAC1?

Q.3.3 Which of the kinase(s) (A-I) have to be knockedout to prevent salicylic acid dependent stomata closure?

Q.3.4 Which site or sites need to be phosphorylated to open the SLAC1 anion channel?
A. S59
B. S120
C. S59 or S120
D. S59 and S120

Q4

Each year, the WHO forecasts which H and N antigens the dominant flu strain is likely to carry to make vaccines. In 2017, an outbreak of strains H1N1 and H3N2 was forecast for the following year. However, in 2018, H1N2 actually became dominant.

Three people (1-3) are vaccinated, and their antibodies are used in immunodiffusion assays. Virus antigens and antibodies are loaded into neighbouring wells of an agar plate according to the table below. They diffuse towards each other and precipitate in a visible band if they crosslink.

Antibody	Well A	Well B
1	H1N1	H1N2
2	H1N1	H3N2
3	H1N1	H1N2

The results of the experiment are presented in Figure 1.

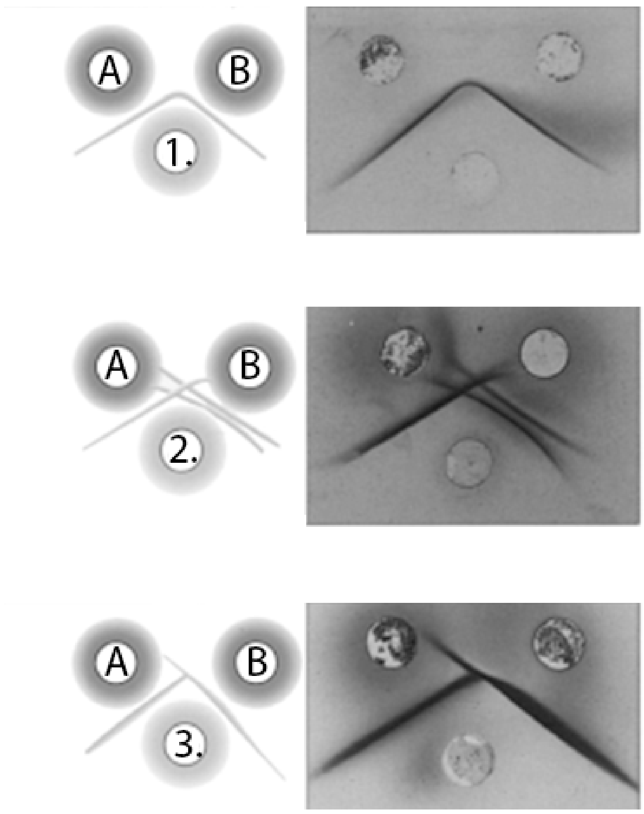


Fig.1

Indicate with an **X** if each the following statements are true (T) or false (F).

Q.4.1	The developed vaccine provided protection for all antigens predicted by the WHO.
Q.4.2	Antibody 1 and 2 came from the same subject.

Q.4.3	The subject that gave Antibody 2 is protected against the new H1N2 influenza outbreak.

Q.4.4	Based on the available information mark antigens that antibody samples (1-3) provide immunity against with an **X**. Mark all other boxes with O.

Catumaxomab is a new anti-cancer antibody, unlike natural antibodies, binds two different antigens, CD3 and EpCAM.

Indicate with an **X** if each the following statements are true (T) or false (F).

Q.4.5	Catumaxomab in an immunodiffusion assay with purified CD3 protein in both antigen wells will give precipitation similar to Antibody 1 in Figure 1.

Q.4.6	The different specificity chains of Catumaxomab are held together by covalent bonds.

Q5

In 1963, a series of experiments were performed by John Cairns and peers to test the mechanism of DNA replication. An experimental organism with circular dsDNA chromosomes was grown for several generations in a medium containing radiolabelled [^3H] thymidine. The organism used these bases to replicate their circular dsDNA chromosomes and therefore the subsequently synthesised DNA became radiolabelled. Samples were rapidly frozen during DNA synthesis and then observed so radiolabelled bases appeared as a line of dark beads on a string in a radiographic picture. They found that each circular chromosome contains a single "eye" during replication, resembling the Greek letter θ (theta structures, Figure 1).

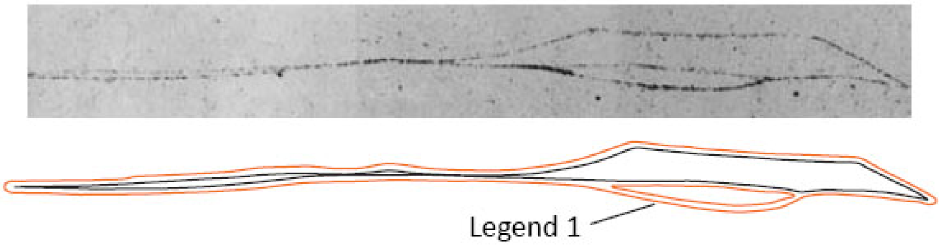


Fig.1. Radiograph and its interpretive drawing of a replicating chromosome. Legend 1 = Replication eye

In a second experiment, researchers grew the cells in a medium that contained both non-labelled and [^3H] thymidine but the latter was only present at low concentrations. Then, during the replicative period, they increased the concentration of [^3H] thymidine in the medium and immediately isolated samples to freeze and photograph. The results are shown in Figure 2.

Fig.2. Radiograph of the replication forks

Indicate with an **X** if each the following statements are true (T) or false (F).

Q.5.1	The experiments show that there is a single replication origin in in one chromosome in this organism.
Q.5.2	The experiments suggest bidirectional replication.

Q.5.3 Which of the following DNA replication models is consistent with the results above? Indicate with an **X**.
A. Conservative
B. Semi-conservative
C. Dispersive
D. None of the above

Q.5.4 What can you conclude from the results on the relative abundance of thymidine? Indicate with an **X**.
A. Thymine is more abundant in the replication origin(s), than in other area of the DNA.
B. Thymine is less abundant in the replication origin(s), than in other area of the DNA.
C. Thymine is equally abundant in the replication origin(s) and in other area of the DNA.
D. Relative thymine abundancy in the replication origin(s) cannot be inferred from these results without doubt.

Q6

During DNA replication, RNA primers are synthesised, which are then extended by DNA polymerase. On the lagging strand, DNA polymerase is then released, forming an Okazaki fragment, and the process repeats. The lagging strand was immobilised at one end to a flow cell. The other end is attached to a florescent bead. Liquid is moved across the cell to stretch the DNA and the position of the bead is measured and plotted as 'DNA length change'.

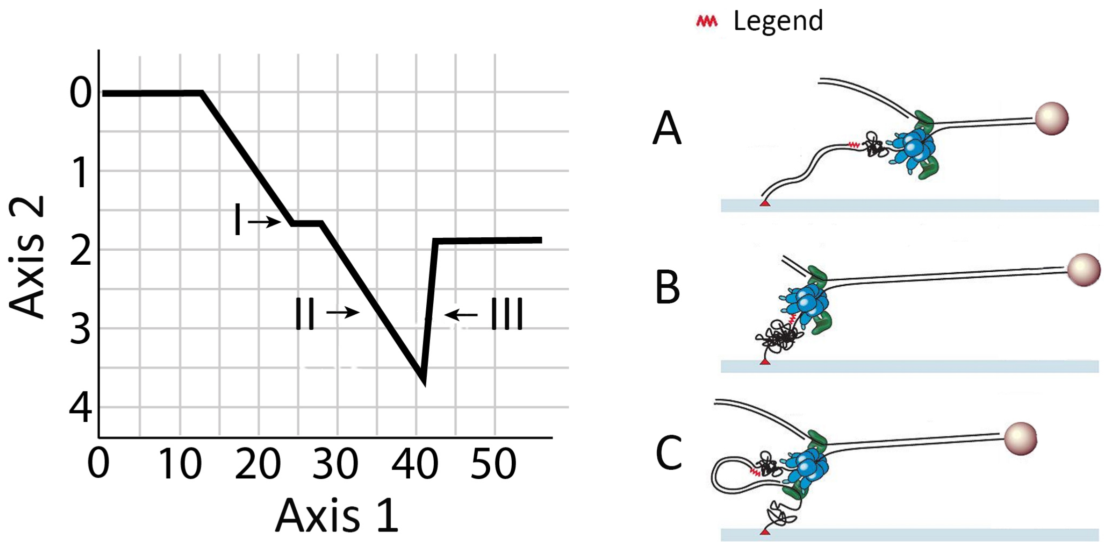


Fig.1. Axis 1 = Time (s). Axis 2 = DNA length change (kb). Legend = Primer.

Q.6.1	Indicate with an **X** which position on the graph (I-III) may correspond to which step of DNA replication (A-C).

Q.6.2	How many base pairs long is the Okazaki fragment based on the figure (to the nearest 500bp)?

Q.6.3	What is the rate of DNA synthesis? (in bp/second)

Q.6.4	1U of Taq polymerase is defined as the amount of enzyme required to incorporate 10nmol dNTP into DNA in 30 minutes under optimal conditions (72°C). In a PCR mixture ($25 \mu L$) made with 1U of Taq pol, assuming unlimited template DNA and optimal conditions throughout, how long would it take theoretically for dNTP levels to drop from a starting $400 \mu M$ concentration to $200 \mu M$ in minutes?

Q7

You find a mutation in p53 ('the guardian of the genome') at nucleotide 42. In your sequence, it is an adenine (A), whereas it is usually guanine (G) in the wild-type, as shown in Figure 1.

5’ … T A A A G T A G A T C T T C C A G A C G C T C T G T T T C C G T … 3’
30 | 40 | 50 |
3’ … A T T T C A T C T A G A A G G T C T G C G A G A C A A A G G C A … 5’

MboII: 5’-GTAGA
BglII: 5’-AGATCT
DpnI: 5’-GATC
HgaI: 5’-GACGC

Fig.1

Q.7.1 You design a test for the mutant allele using restriction enzymes. Indicate with an **X** which enzyme (A-D) can be used to distinguish patient samples?
A. MboII
B. BglII
C. DpnI
D. HgaI

A clinician uses the appropriate enzyme to carry out restriction fragment length polymorphism analysis. Results for patient A-G are shown in Figure 2.

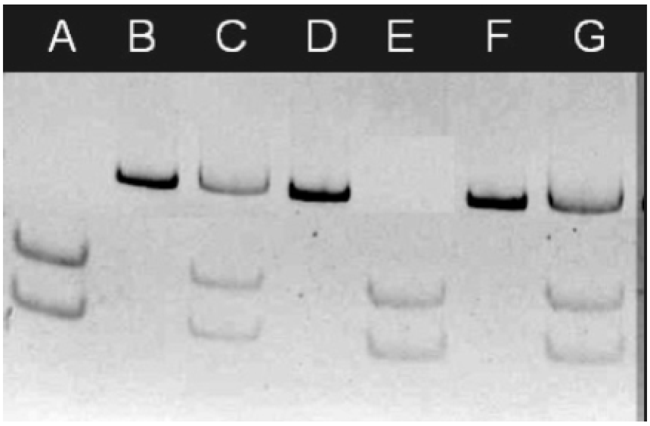


Fig.2

Q.7.2 Determine the genotype of the patients by marking in it in the table with an **X**. The letters in the table refer to the nucleotides found at position 42.

Following genotyping, p53 transcript and protein were measured as seen in Figure 3.

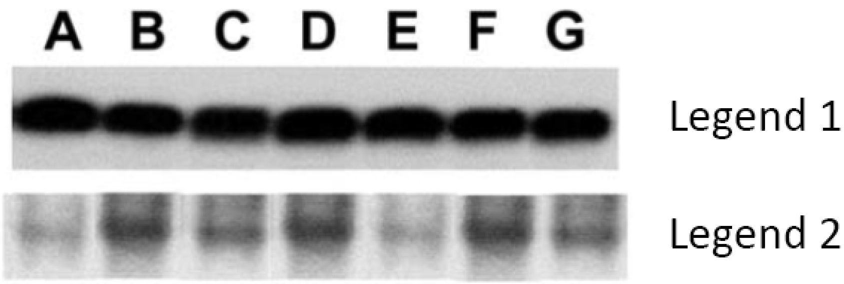


Fig.3. Legend 1 = RNA. Legend 2 = Protein.

Q.7.3 Indicate with an **X** whether the mutation affects:
A. p53 transcription
B. p53 translation
C. Both

Q.7.4 Normally, when the genome is damaged, four molecules of p53 form a complex which can then bind to DNA as a transcription factor. In an experiment a cell is heterozygous for a mutation which produces p53 that prevents DNA binding. If the amount of transcription of p53 target genes is proportional to the level of active p53, what fraction of transcription occurs in this cell compared to normal cells?

Animal Anatomy and Physiology

Q8

In an experiment, the breathing tube of the spirometer was filled with oxygen and fitted with a carbon dioxide absorber that removed all exhaled CO_2, as illustrated in Figure 1. Test subject "A" was allowed to breathe from the apparatus for 4 minutes. The result of this experiment is presented in Figure 2.

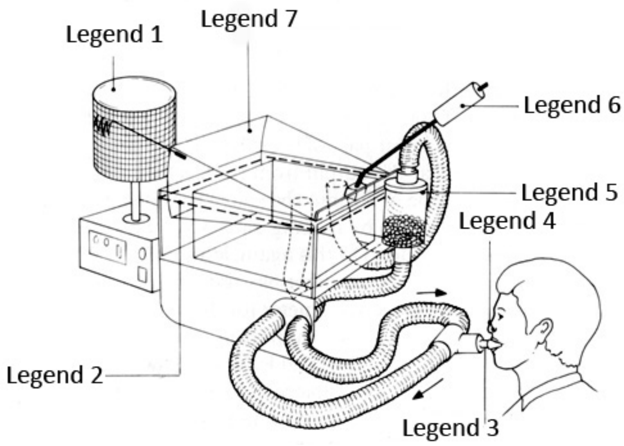


Fig.1. Legend 1 = Kymograph (graphical registry of mechanical movement over time). Legend 2 = Water level. Legend 3 = Mouthpiece. Legend 4 = Nose clip. Legend 5 = Carbon dioxide absorber. Legend 6 = Counterpoise. Legend 7 = Spirometer chamber.

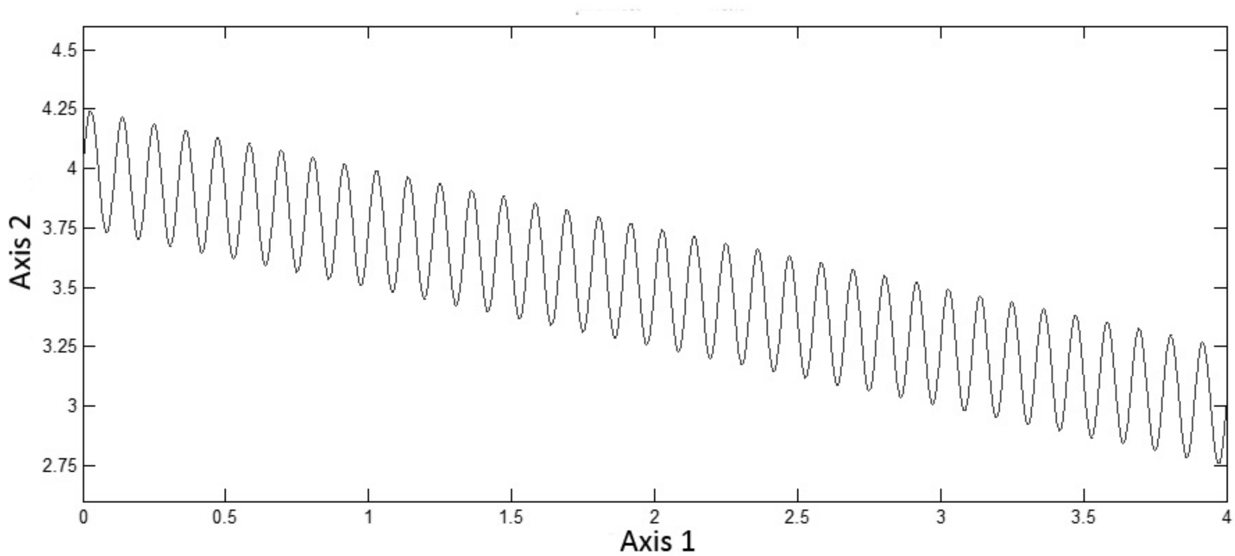


Fig.2. Spirometer experiment. Axis 1 = Time (min). Axis 2 = Spirometer volume (L).

Q.8.1	What is the tidal volume of test subject "A"? (in mL)

Q.8.2	What is the breathing rate of test subject "A"? (in breaths/min)

Q.8.3	What is the oxygen consumption of test subject "A"? (in mL/min)

Q.8.4	In an experiment where the CO_2 absorber is removed, how would the test subject's breathing frequency change? Indicate your answer by putting an X in the appropriate box on the answer sheet. A. Increase B. Not change C. Decrease

In a different spirometry experiment, we measured the volume and speed of air during normal and forced exhalation and inhalation.

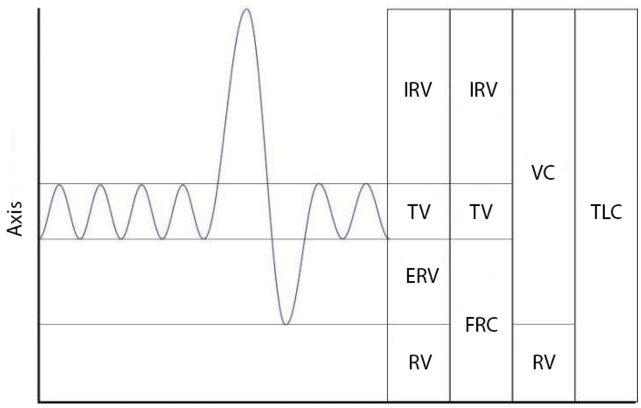


Fig.3. Axis = Volume (ml/kg).

- TLC - Total lung capacity: the volume in the lungs at maximal inflation, the sum of VC and RV.
- TV - Tidal volume: that volume of air moved into or out of the lungs during quiet breathing
- RV - Residual volume: the volume of air remaining in the lungs after a maximal exhalation
- ERV - Expiratory reserve volume: the maximal volume of air that can be exhaled from the end-expiratory position
- IRV - Inspiratory reserve volume: the maximal volume that can be inhaled from the end-inspiratory level
- VC - Vital capacity: the volume of air breathed out after the deepest inhalation, followed by a full expiration.
- FRC - Functional residual capacity: the volume in the lungs at the end-expiratory position

Consider test subject "B" who had the following results:

- VC = 5,200 mL
- IRV = 3,300 mL
- FRC = 2,500 mL
- RV = 1,300 mL

Q.8.5	Calculate the tidal volume of test subject "B" and give your answer on the answer sheet in millilitres (mL).

Test subject "C" was also tested. After several measurements, researchers found that her tidal volume is 500 mL and respiratory rate is 15/min when resting. During intense exercise her O_2 consumption rises leading to a 3-fold higher volume of air inhaled per minute and an increased respiratory rate of 25/min.

Q.8.6 Based on the graph above and the data provided, calculate the tidal volume of test subject "C" during exercise. Give your answer on the answer sheet in millilitres (mL).

Q9

In the brainstem there is a neuronal regulation network which maintains the rhythmic breathing as it can be seen in the list below.

Pneumotaxic center (PC):

- Controls the duration of inhalation by signalling to inspiratory area
- Strong activity $\longrightarrow$ length of inspiration < 0.5 seconds
- Weak activity $\longrightarrow$ length of inspiration > 5 seconds

Apneustic center (AC):

- Promotes inhalation by stimulating the DRG
- Inhibited by pulmonary stretch receptors
- Inhibits PC

Dorsal respiratory group (DRG):

- Controls the basic rhythm by triggering respiratory impulses
- Vagal and glossopharyngeal nerves bring peripheral chemosensory information to it

Ventral respiratory group (VRG):

- Inactive during normal baseline respiration
- High DRG activity (increased need for ventilation) stimulates it
- Has neurons for both inhalation and exhalation
- Inhibits AC

Central chemosensitive area (CC):

- Stimulated by high blood pCO_2
- Excites other respiratory centres

Decide how the described features would change (increase, no change or decrease) in the following scenarios. Indicate your answer by putting an X in the appropriate box on the answer sheet.

- A. Increase
- B. No change
- C. Decrease

Q.9.1	Respiratory rate upon damage to pneumotaxic center

Q.9.2	Blood pH after a stroke affecting central chemosensitive area neurons

Q.9.3	Ventral respiratory group activity upon a deep inhalation

Q.9.4	Central chemosensitive area neuron membrane hyperpolarisation upon voluntarily holding one's breath

Q.9.5	Length of inhalation upon overstimulation of apneostic center

Q10

Pressure–volume work (or PV work) occurs when the volume of a system changes. It can be calculated as the product of Δ volume and Δ pressure. The work of breathing is mainly determined by the compliance of the lungs, which is defined as:

Compliance = Δ volume (mL) /Δ pressure (cm H_2O)

Compliance itself is dependent on two main factors, (1) the elastic characteristics of the lung, and (2) the surface tension generated by the air-liquid interface at the inner alveolar surface. In 1929, von Neergaard excised the lung of a cat which he then inflated and then deflated using air and then inflated and deflated using saline solution. The results are plotted in the figure below.

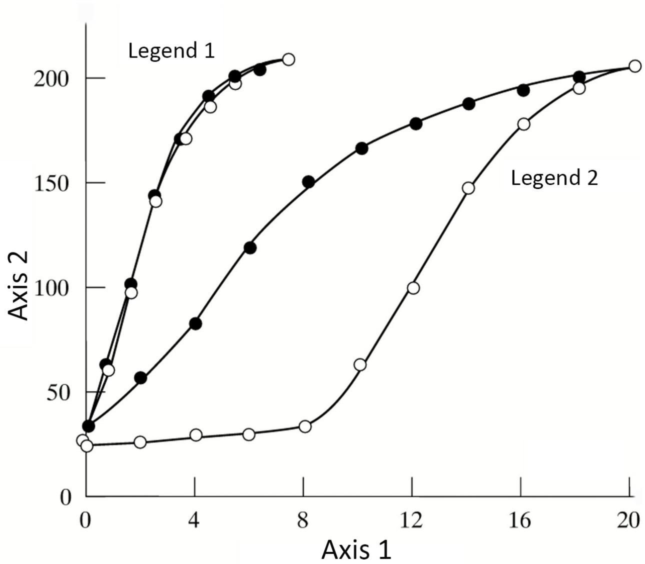


Fig.1. Axis 1 = Pressure (cm H_2O). Axis 2 = Volume (ml). Legend 1 = Saline inflation. Legend 2 = Air inflation. Empty circles ◦ : inflation. Full dots • : deflation.

For the following pairs of statements (A and B), evaluate the mathematical relationship between the values. Indicate your answers using the following symbols > or < or = in the cells provided on the answer sheet.

Q.10.1 A: Work of inflating the cat lung with saline solution.
B: Work of inflating the cat lung with air.

Q.10.2 A: The compliance of the lung at volumes above 35 mL (upon normal air inhalation).
B: The compliance of the lung at volumes below 35 mL (upon normal air inhalation).

Q.10.3 A: Contribution of intrinsic elastic forces to total lung compliance.
B: Contribution of surface tension to total lung compliance.

Q.10.4 A: Compliance of a healthy, normally functioning lung.
B: Compliance of a lung in which surfactant production is abolished by genetic mutations.

Q.10.5 A: Total lung capacity in a normally functioning cat lung.
B: Total lung capacity in the same cat's lung after chemical inhibition of surfactant production.

Q11

The electromyogram (EMG) is a technique used to measure the collective electrical activity of active skeletal muscle fibres using electrodes placed on the skin. A human subject was prepared for a muscle electrophysiology experiment to test the conduction velocity of their motor neurons. The experimental setup and a trace of one of their measurements are illustrated below.

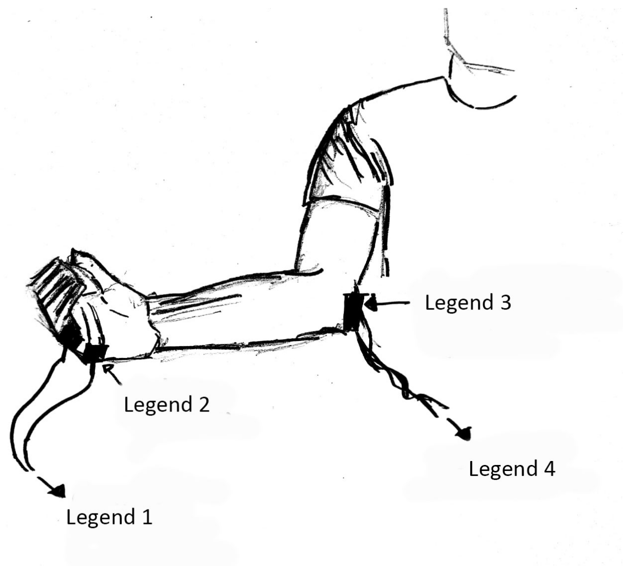


Fig.1. Legend 1 = Connected to amplifier and detector. Legend 2 = Stick-on electrodes. Legend 3 = Stimulator applied to notch in back of elbow. Legend 4 = To isolated stimulator

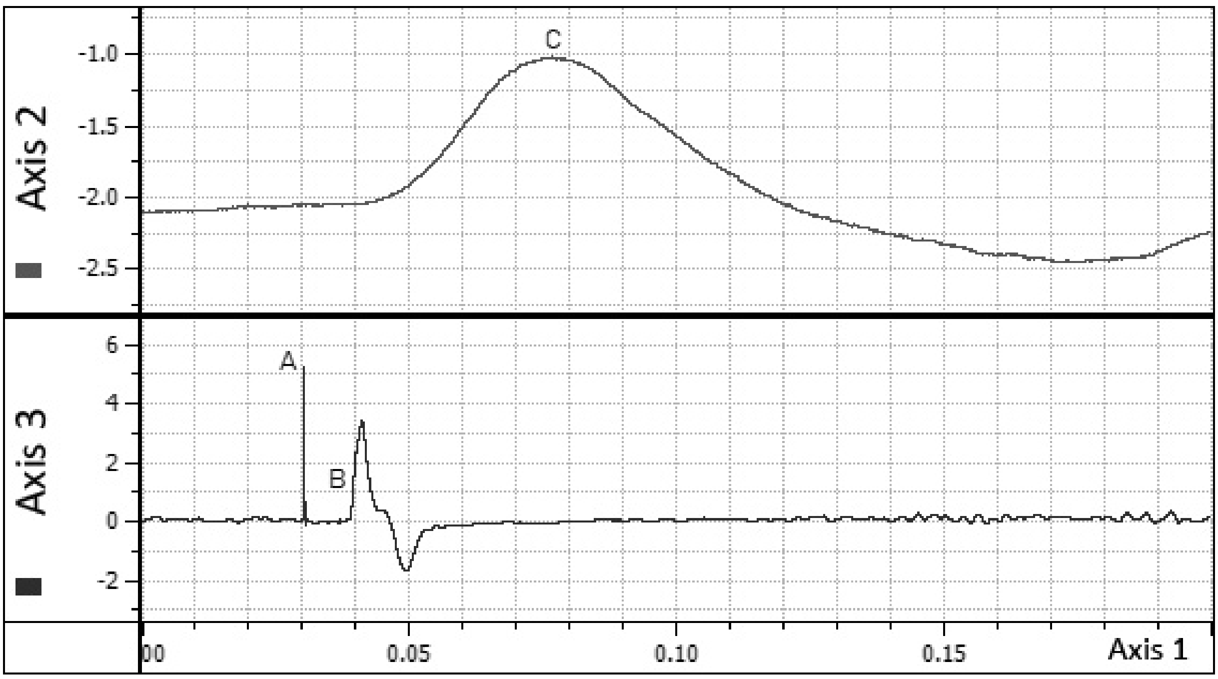


Fig.2. Axis 1 = Time (sec). Axis 2 = Force transducer (N). Axis 3 = Electromyogram (mV).
(The spike in the "Electromyogram" trace at "A" (time = 0.03 sec) is added by the computer to show when the stimulus was presented.)

Q.11.1 Calculate the conduction velocity of the motor neuron. Assume that the initiation and transmission of signalling within the muscle cells takes zero milliseconds. The distance between the stimulation electrode and the EMG electrode was 420 mm. Give your answer in $m \times \sec^{-1}$ and round it to the nearest whole number.

For the following questions, answer with the letters of the processes listed below (A-I). Note that not all of the options are actually involved in the illustrated process. Indicate your answer by putting an X in the appropriate box on the answer sheet.

A. Ca^{2+} channel opening in the membrane of the sarcoplasmic reticulum

B. Acetylcholine released

C. Ca^{2+} binding to troponin

D. Action potential initiated in muscle fibre membrane

E. Action potential propagated along motor neuron

F. Actin-myosin cross-bridge cycling

G. Troponin blockage of myosin binding sites restored

H. Active transport of Ca^{2+} across membrane of sarcoplasmic reticulum

I. Ca^{2+} concentration increases in the muscle fibre's cytoplasm

Q.11.2 Which **2** processes happen between time points A and B?

Q.11.3 What happens immediately at point B?

Q.11.4 Which process accounts for the observation of the peak labelled C?

Q.11.5 Which process is most likely to account for the slowness of the fall of the trace after point C?

Q12

Nitric oxide (NO) plays a role in physiological regulation of the circulatory system. NO is the activator of soluble guanylyl cyclase, leading to the formation of cyclic GMP (cGMP), an important second messenger in vascular smooth muscle.

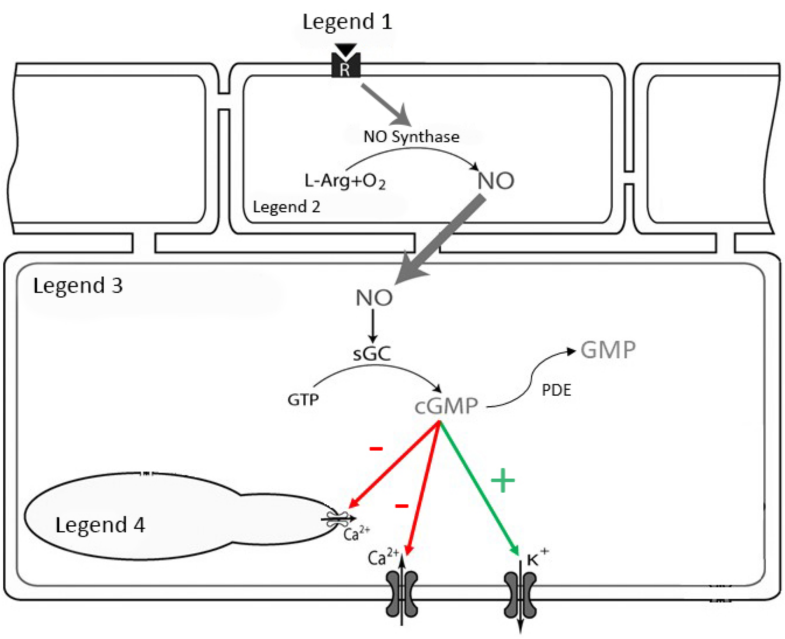


Fig.1. Legend 1 = Adrenaline. Legend 2 = Endothelial Cell. Legend 3 = Smooth Muscle Cell. Legend 4 = Sarcoplasmic Reticulum.

The contraction of vascular smooth muscle determines the diameter of a blood vessel. The relationship between vessel diameter, pressure, and flow is described by Poisseuille's law.

$$Q = \frac{\pi r^4 \Delta P}{8 \mu L}$$

Where:

- Q is flow in mL/min
- r is the vessel radius
- ΔP is the pressure gradient
- μ is the viscosity of the fluid in the vessel
- L is the length of the vessel

Based on the above, how would the specified characteristics be changed in the following scenarios? For the **next three questions**, decide if the result is increase, no change or decrease. Indicate your answer by putting an X in the appropriate box on the answer sheet.

A. Increase

B. No change

C. Decrease

Q.12.1 Flow in the vessel in case of fight-or-flight response in a living organism.

Q.12.2 Flow in the vessel upon administration of the drug sildenafil, which inhibits cGMP specific phosphodiesterase (PDE).

Q.12.3 Membrane potential of the smooth muscle cell upon adrenaline infusion into a perfused vessel preparation.

Q.12.4 By how many fold must pressure increase, to create an increase in flow equivalent to a 10-fold increase in vessel radius? Provide your answer in the appropriate box on the answer sheet.

Q13

The water content of the human body is about 60%. The distribution of this huge amount of fluid and their content are both important to understand the main processes of homeostasis. Figure 1 shows the amount of fluid in, and the ionic content of, different body compartments.

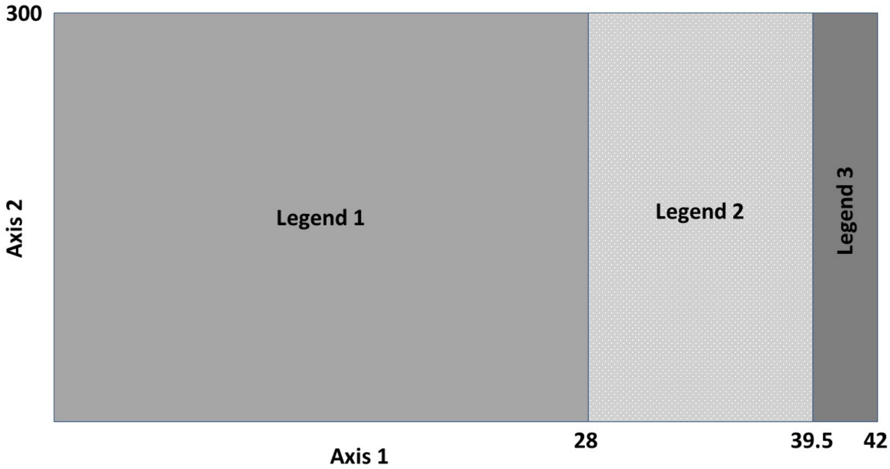


Fig.1. Distribution of the different fluid compartments in a healthy adult. Axis 1 = Mass (kg). Axis 2 = Osmotic concentration (mOsmol/kg). Legend 1 = Intracellular fluid. Legend 2 = Interstitial fluid. Legend 3 = Blood plasma.

Inadequate supply of water and ions or the abnormal loss of these substances alter the values found in Figure 1. In Figure 2, four physiological conditions with their distinctive values are shown, in which changes occur in the mass and/or osmotic concentration of body fluids. The diagrams only show the intracellular (dark grey) and the extracellular (light grey) fluid compartments and their mass and osmotic concentration values. The areas surrounded by the dashed lines represent the normal values of the compartments shown in the previous diagram.

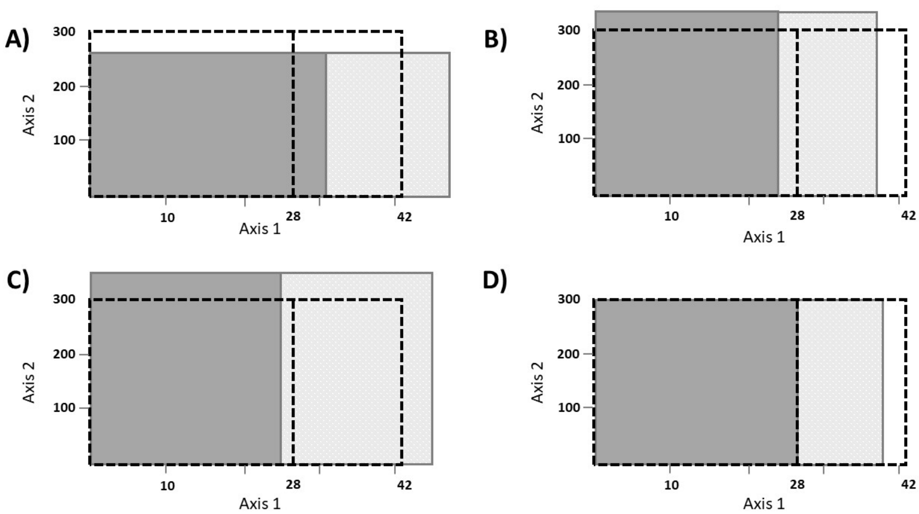


Fig.2. Axis 1 = Mass (kg). Axis 2 = Osmotic concentration (mOsmol/kg)

Q.13.1 Calculate the extracellular fluid mass / intracellular fluid mass ratio of sodium ions, using normal (healthy) values. Assume that the concentration of sodium ions is 140 mOsm/kg in extracellular fluid, and 10 mOsm/kg in intracellular fluid. Give your answer to the nearest integer (whole number).

Match the following statements describing different physiological conditions with the letters in Figure 2 (A-D). Indicate your answer by putting an X in the appropriate box (A-D) on the answer sheet.

Q.13.2 A condition typical after drinking large quantities of diluted liquid.

Q.13.3 A condition typical of dehydration (loss of water).

Q.13.4 A condition brought about after severe vomiting and diarrhoea.

Q.13.5 A condition most likely to cause the blood pressure to increase.

Q.13.6 Which of these conditions will inhibit release of vasopressin (ADH) into the blood?

Q14

Insulin is a 51-amino acid protein consisting of 2 polypeptide chains linked by disulphide bonds.

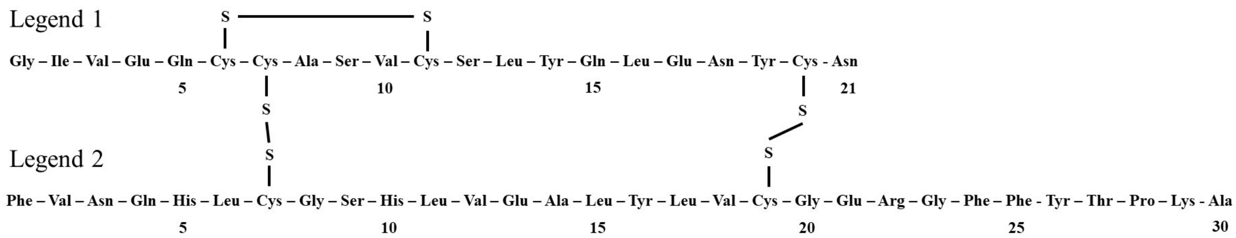


Fig.1. Legend 1 = A chain. Legend 2 = B chain

In type 1 diabetes mellitus, there is no adequate insulin release due to the loss of beta-cells of the pancreas. Therefore, the only treatment is insulin injection, which is most commonly administered into the fat under the skin. Here it forms hexamers, which then dissociate to be absorbed into the blood stream.

The graph shows the plasma level of three available insulins (1-3) after subcutaneous injection:

1. Human insulin produced in *Saccharomyces cerevisiae* by recombinant DNA technology.

2. An insulin analogue in which swapping one amino acid reduces dimer and hexamer formation.

3. An insulin analogue in which amino acid changes to the A and B chains change the molecule's isoelectric point from 5.7 to a more neutral pH.

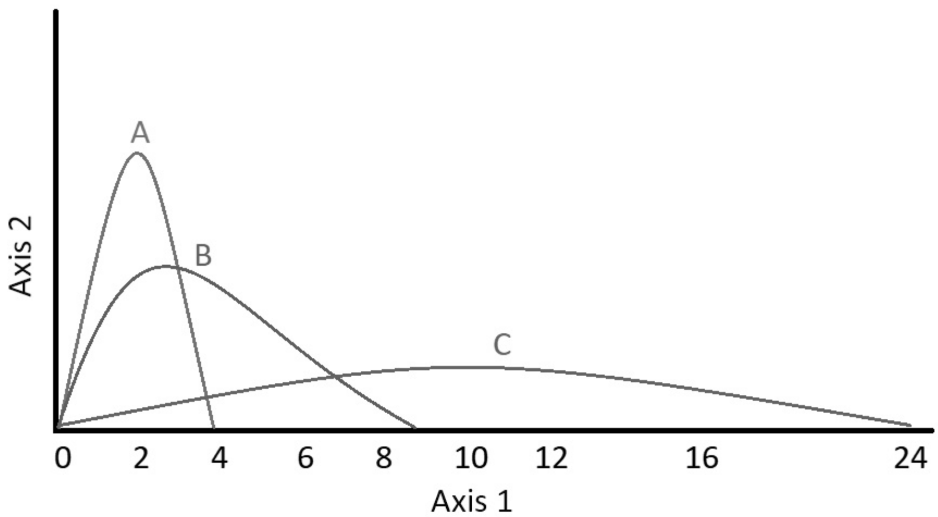


Fig.2. Axis 1 = Time. Axis 2 = Relative plasma insulin level

Q.14.1 Match the insulin types (1-3) with the plots (A-C) based on the information above. Indicate your answer by putting an X in the appropriate box on the answer sheet.

Indicate with an **X** if each of the following statements is true (T) or false (F).

Q.14.2 Oral administration (e.g. in the form of pills) of insulin is approximately as effective as subcutaneous administration.

Q.14.3 Stimulation of insulin release from pancreatic beta cells can be an alternative therapy in type 1 diabetes mellitus.

Q.14.4 Immunosuppressant therapy could slow down the onset of type 1 diabetes mellitus if it is diagnosed early.

Q.14.5 The long-acting, subcutaneously administered insulins (e.g. plot C) act longer because their degradation in the circulation is slower.

Q15

An audiometry exam tests a subject's ability to hear sounds at different frequencies. Sound waves travel to the inner ear through the ear canal, eardrum, and the bones of the middle ear (air conduction, empty circles (○) or alternatively, through the bones around and behind the ear (bone conduction, full dots (●).

Figure 1. shows the lowest sound intensity (in dB) at a given frequency (Hz) the test subject can still perceive, where a 0 value is the population average and any positive value suggests that a higher than average intensity is required for the subject to hear the sound.

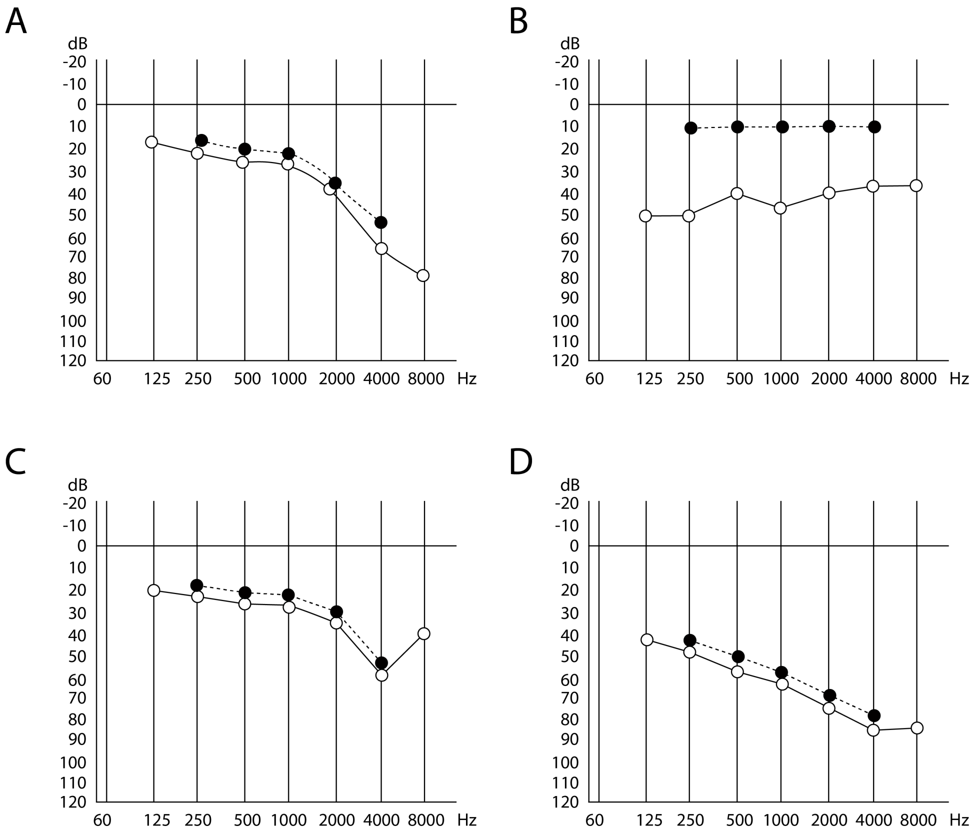


Fig.1

Q.15.1 What is the highest frequency at which bone conduction is tested? (Hz)

Match the following conditions with the test results in Figure 1 (A-D). Indicate your answer by putting an X in the appropriate box on the answer sheet.

Q.15.2 Old age related reduced sensitivity to high frequency.

Q.15.3 Middle ear infection impeding the movement of the auditory bones.

Q.15.4 Neurological hearing damage causing overall reduction in hearing acuity.

Q.15.5 Damage caused by an air horn.

Q16

In eutherians (placental mammals), the developing fetus and the placenta are surrounded by two membranes, the chorion and the amnion. In monozygotic twin pregnancies, these membranes can be either separate or shared. This depends on the developmental stage at which the initial conceptus splits into two separate ones.

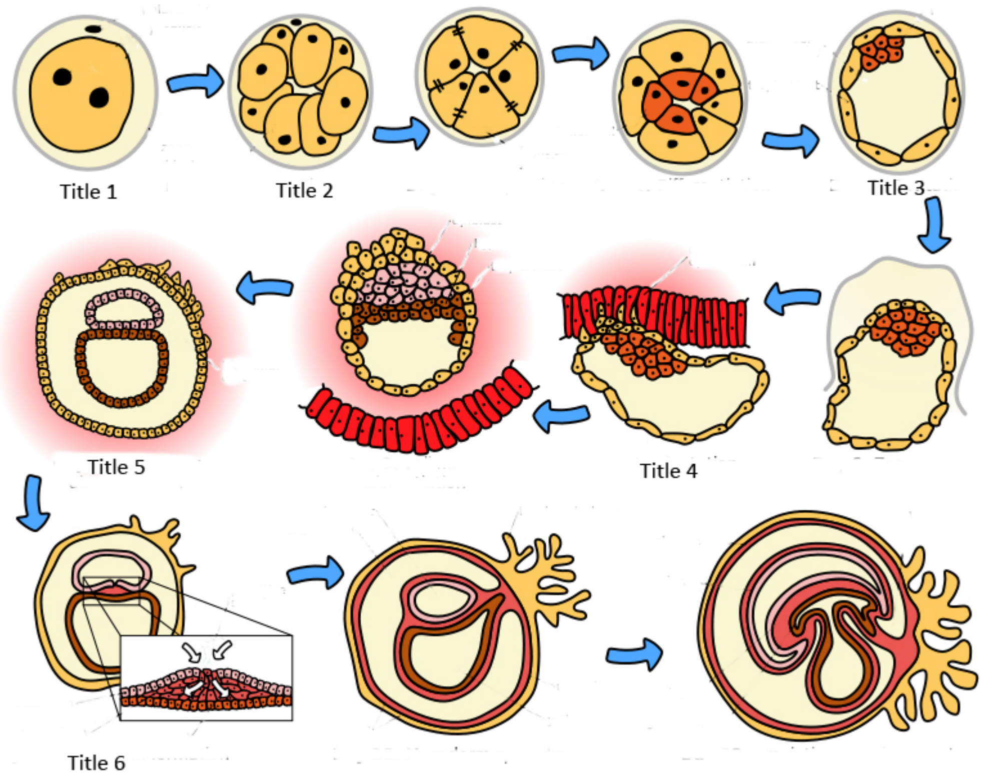


Fig.1. Title 1 = Day 1 Fertilization. Title 2 = Day 2 Morula formation. Title 3 = Day 5 Blastocyst formation. Title 4 = Day 7 Implantation. Title 5 = Day 9 Inner cell mass differentiation. Title 6 = Day 12 Germ disk and then primitive streak formation.

In each of the following scenarios (Q.16.1-4), decide if the chorion and the amnion are shared or separate. Indicate your answer by putting an X in the appropriate box (A-D) on the answer sheet.

Your options:

A. Both chorion and amnion are shared

B. Chorion is separate, amnion is shared

C. Chorion is shared, amnion is separate

D. Both chorion and amnion are separate

Q.16.1 Dizygotic twins

Q.16.2 Monozygotic twins where splitting occurred in the morula stage

Q.16.3 Monozygotic twins where splitting occurred in the inner cell mass stage

Q.16.4 Monozygotic twins where splitting occurred between the formation of the germ disc and the primitive streak

Plant Anatomy and Physiology

Q17

Nutrient availability is thought to be one of the most important conditions influencing stem structure. It is believed to induce changes in the structure of aerenchyma (a spongy tissue of air channels) and the distribution of lacunae (airspaces in aerenchyma tissue). To investigate this, biomechanical parameters of stems of two aquatic plant species (*M. s. = M. scorpoides and M. a. = M. aquatica*) were measured under low (↓ in table) or high (↑ in table) nutrient conditions. Data were statistically evaluated and the mathematical relationships between these parameters determined as shown in Table1. (ns = non-significant, more stars show greater significant difference).

	M.s. ↓	M.s. ↑	Sign.	M.a. ↓	M.a. ↑	Sign.
I (mm^4)	0.57±0.27	0.47±0.18	ns	2.68±1.95	3.40±2.30	ns
Icor(mm^4)	0.47±0.18	0.39±0.15	ns	2.39±1.63	2.96±2.11	ns
BF (N)	2.09±0.62	0.77±0.83	***	5.35±2.03	5.98±2.17	*

Table 1. M.s. ↓ = *M. scorpoides* in low nutrient conditions. M.s. ↑ = *M. scorpoides* in high nutrient conditions. Sign. = Statistical significance. M.a. ↓ = *M. aquatica* in low nutrient conditions. M.a. ↑ = *M. aquatica* in high nutrient conditions. BF = Breaking force.

$$I(M.s.) = \frac{\pi R^4}{4} \tag{1}$$

$$I(M.a.) = \frac{L^4}{12} \tag{2}$$

$$Icor = I(1-P) \tag{3}$$

Where:

- **Second moment of area** (I) = describes the geometry of the stem cross section
- **Breaking force** (BF) = the maximum force (F) the stem fragment could withstand before breaking
- **Porosity** (P) = the ratio of total lacunar area and total cross-sectional area of stem
- **Icor** = I corrected by porosity
- **Radius** of a circular stem cross section (R)
- **Side length** of a rectangular stem cross section (L)

Calculate the mean porosity of the stems of both plant species for both nutrient conditions using two significant figures for your result.

Q.17.1 Mean porosity of M.s. in low nutrient conditions

Q.17.2 Mean porosity of M.s. in high nutrient conditions

Q.17.3 Mean porosity of M.a. in low nutrient conditions

Q.17.4 Mean porosity of M.a. in high nutrient conditions

Representative cross sections of the stems of M.s. and M.a. are shown below.

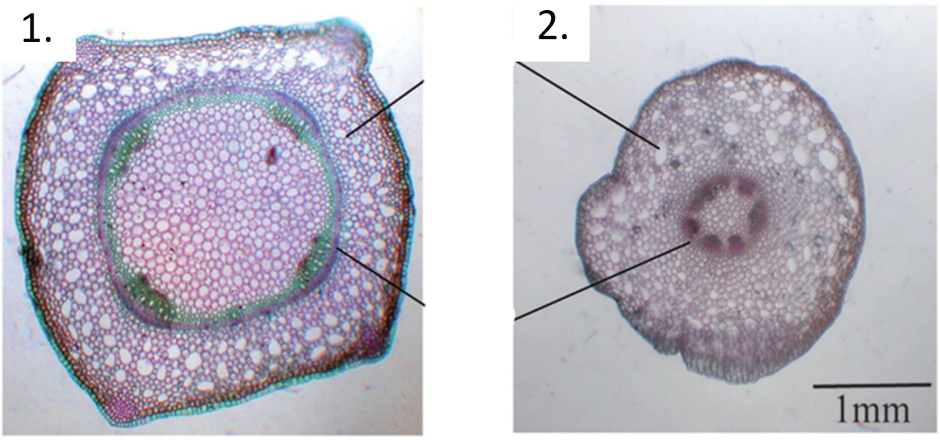


Fig.1

Q.17.5 Match the following cross sections (1 and 2) with the two species (A and B, as specified below). Indicate your answer by putting an X in the appropriate box on the answer sheet.
A. if the species is *M. scorpoides*, and B. if the species is *M. aquatica*.

Q.17.6 In a sudden flood, which M. s. plants are more likely to be damaged? Indicate your answer by putting an **X** in the appropriate box on the answer sheet.
A. Growing on high nutrient soil.
B. Growing on low nutrient soil.

Q18

Two transcription factors, GNC and GLK, have been implicated in regulating chloroplast development. Three mutants with different combinations of knock-out mutations in these genes were created in *A. thaliana* as follows: *GNC* or *GLK* only and *GLC-GNK* with mutations in both genes. These mutants and the wild type plant (WT) were then analysed with respect to various parameters associated with chloroplast development as shown in Figure 1. (different letters indicate significantly different values)

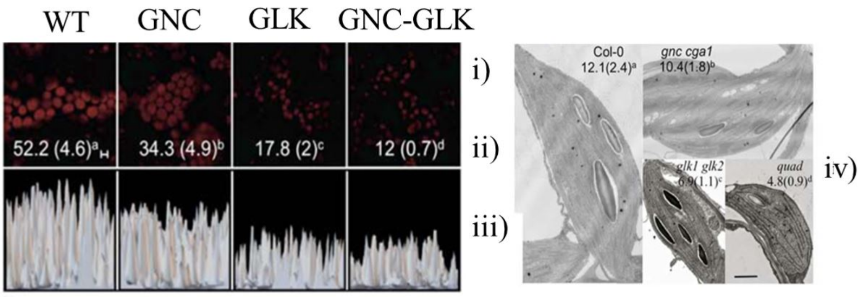


Fig.1. i) Average chloroplast fluorescence. ii) Chlorophyll a content (ng/mm2 $\pm$ SD). iii) Chloroplast relative fluorescence intensities. iv) Microscopic images of chloroplasts

Q.18.1 Select the microscopic technique used to take the images shown in Figure 1.i and Figure 1.iv. Indicate your answer by putting an X in the appropriate box on the answer sheet.

Microscopic techniques:
A. Bright-field light microscopy
B. Fluorescence microscopy
C. X-ray microscopy
D. Atomic force microscopy
E. Transmission electron microscopy
F. Scanning electron microscopy

Q.18.2 Which of the GLK and GNC transcription factor families is the stronger positive regulator of chloroplast development? Indicate your answer using the symbols, > or < or = in the appropriate box on the answer sheet.

Q.18.3 Select the single best statement describing the contribution of the GLK and GNC genes to chlorophyll biosynthesis. Indicate your answer by putting an X in the appropriate box on the answer sheet.

A. The GLK and GNC genes modify the expression of chlorophyll biosynthetic genes in separate pathways for chlorophyll biosynthesis.
B. GLK genes modify the expression of chlorophyll biosynthetic genes more upstream than the chlorophyll biosynthetic genes controlled by GNC.
C. GNC genes modify the expression of chlorophyll biosynthetic genes more upstream than the chlorophyll biosynthetic genes controlled by GLK.
D. The relative contribution of the GLK and GNC regulated chlorophyll biosynthetic genes cannot be determined based on the data above.

Q19

Two transcription factors, GNC and GLK, have been implicated in regulating chloroplast development. Three mutants, GNC single-, GLK single-, and GNC-GLK double mutants were analysed for the expression of nine genes (PSAE-2, PSAF, PSBY,CHLM, LHCA2, LHCB3, rbcL, psaC and psbA) using reverse transcription quantitative PCR (RT-qPCR).

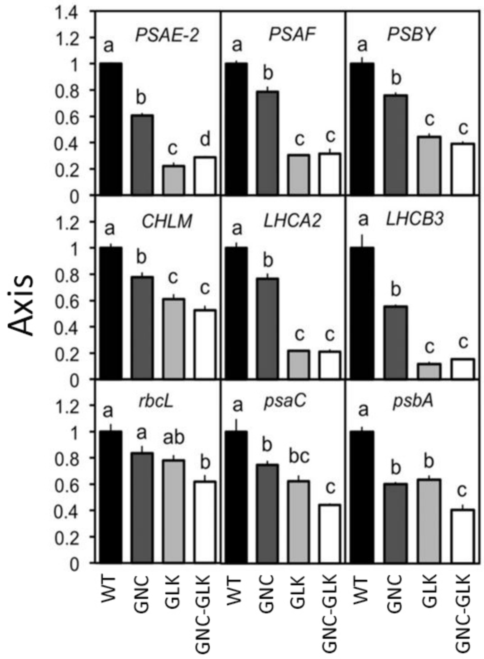


Fig.1. Axis = Relative expression. Note that different letters indicate significantly different values.

Q.19.1 Arrange the following experimental steps in the appropriate order and fill in the flowchart to summarise the workflow of a RT-qPCR. Put one letter in each box. You do not have to use all the letters. We already put letter A in the correct position for you.

- ***A. Reverse transcribe into cDNA (complementary DNA) using reverse transcriptase***
- B. Add primers of random sequence and DNA-dependent DNA polymerase
- C. Analyse the results by determining the number of PCR cycles needed for each sample to reach a threshold fluorescence and compare these to one another
- D. Take plant tissue samples
- E. Purify total mRNA
- F. Add a sequence-specific probe containing a quencher and a fluorophore, primers for the sequence of interest and DNA-dependent DNA polymerase
- G. Measure the fluorescence intensity at the end of the PCR amplification with a spectrophotometer
- H. Purify total DNA with RNase
- I. Incubate in a thermocycler under the appropriate conditions to allow for the amplification of the sequence of interest and the release of the fluorophore from the probe. Simultaneously, computationally quantify the increase in fluorescence due to the fluorophore release
- K. Create cell lysate

Q.19.2 Given that GNC and GLK act as transcription factors, determine if they are positive or negative regulators of each of the the nine target genes (PSAE-2, PSAF, PSBY,CHLM, LHCA2, LHCB3, rbcL, psaC and psbA). Select the genes that are **upregulated** by GNC and GLK and indicate your answer by putting X in the appropriate boxes on the answer sheet.

Q20

The following question concerns the electron transport chain of photosynthesis. Note that NADP+/NADPH is at a higher free energy state than the manganese cluster of the oxygen evolving complex but lower than the central chlorophyll pair of photosystem II in the excited state.

Q.20.1 Put the following oxidised/reduced pairs in an order of increasing redox potential. Put one letter in each box on your answer sheet. We already put letter F in the correct box.

- A. Oxidised/reduced central chlorophyll pair of photosystem I in the excited state
- B. $NADP^{+}$ or $NADPH$
- C. Fully reduced/fully oxidised manganese cluster of the oxygen evolving complex
- D. Oxidised/reduced central chlorophyll pair of photosystem II in the ground state
- E. Oxidised/reduced central chlorophyll pair of photosystem II in the excited state
- ***F. Oxidised/reduced electron carrier Tyr residue***

Q21

Not all flowers use the ABC model of genetic regulation. *Nigella* species have spiral flowers and variable numbers of sepals (C), petals (D), stamens (E) and carpels (F) under the control of a genetic system different from that of ABC. In an experiment (Fig.1) Genes involved in flowering (leftmost column) were silenced and their morphology was assessed. The number of each organ (C-F) was measured and results are indicated in the bar charts (right column). Bar G shows the sum of all the numbers (C-F).

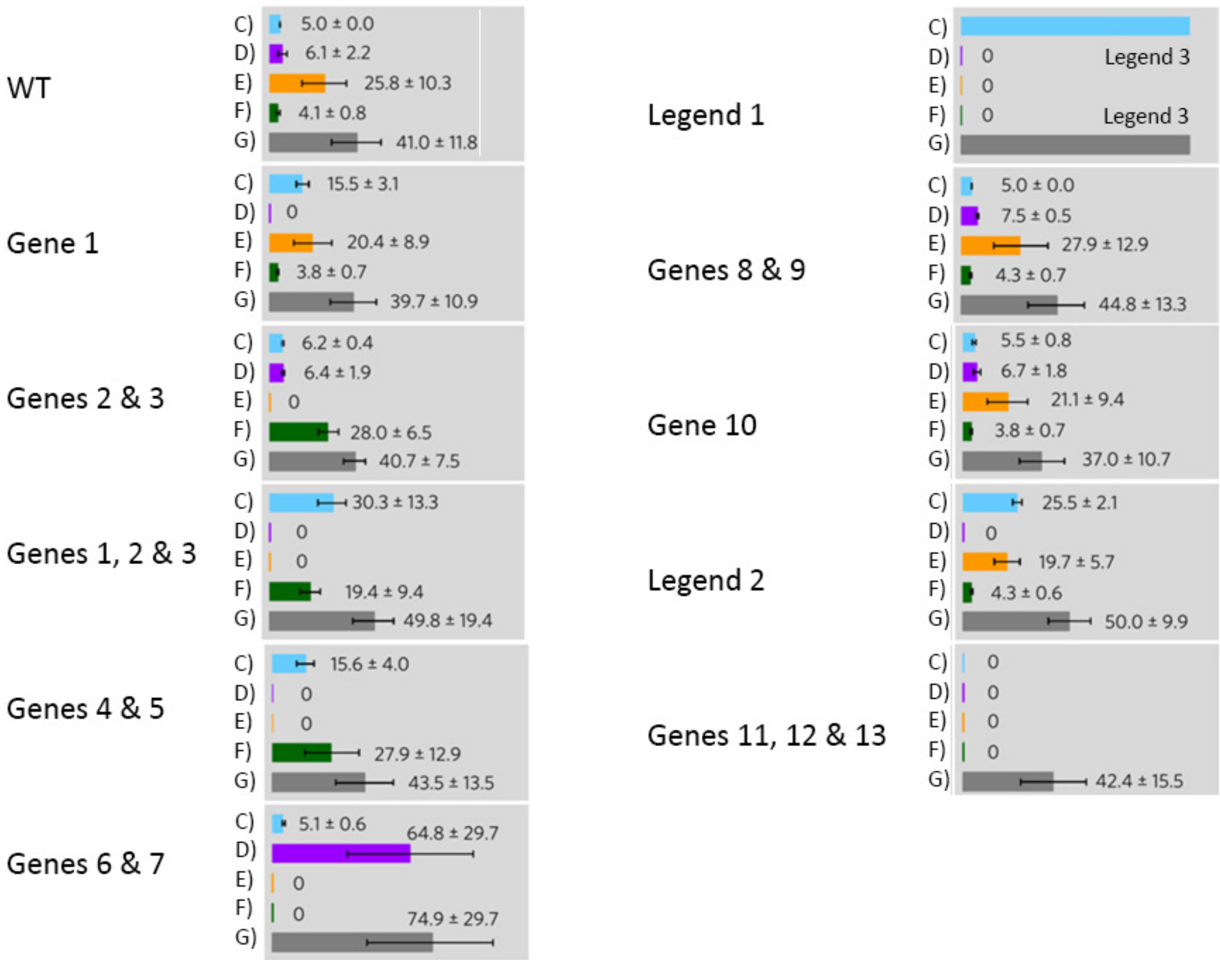
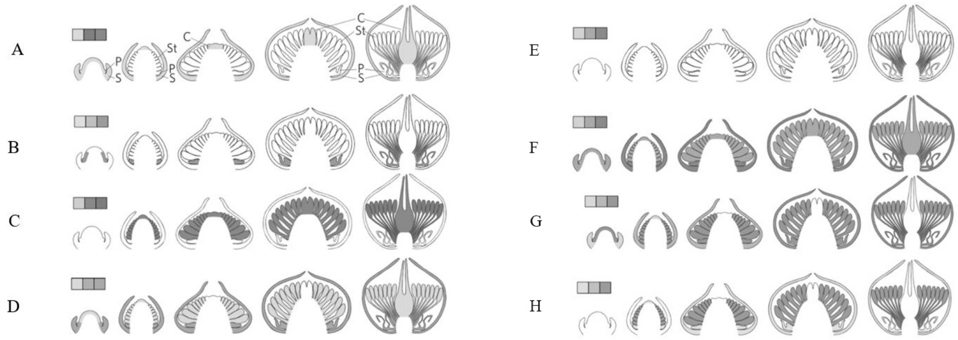


Fig.1. WT = Wild-type. Legend 1 = Naturally occurring gene 1 KO mutant + genes 6 and 7 silenced. Legend 2 = Naturally occurring gene 1 KO mutant + gene 10 silenced. Legend 3 = Countless

Q.21.1 Each gene is expressed according to one of the following patterns (darker = more expression). Match the genes (1-13) with the patterns (A-H). Indicate your answer with an **X** in the appropriate boxes on the answer sheet. Each letter is only used once!



Q22

Venation and size of leaves are important factors influencing plant physiology. Veins can be categorised hierarchically into four groups: midvein (1°), secondary (2°), tertiary (3°) and quaternary (4° or minor veins) as indicated by arrows on Figure 1. Scaling between vein density (= vein length per total leaf area) and leaf area as well as vein diameter and leaf area were determined for 500 species of dicotyledons. [● = data from species with branched veins; ◦ = data from species with parallel veins]

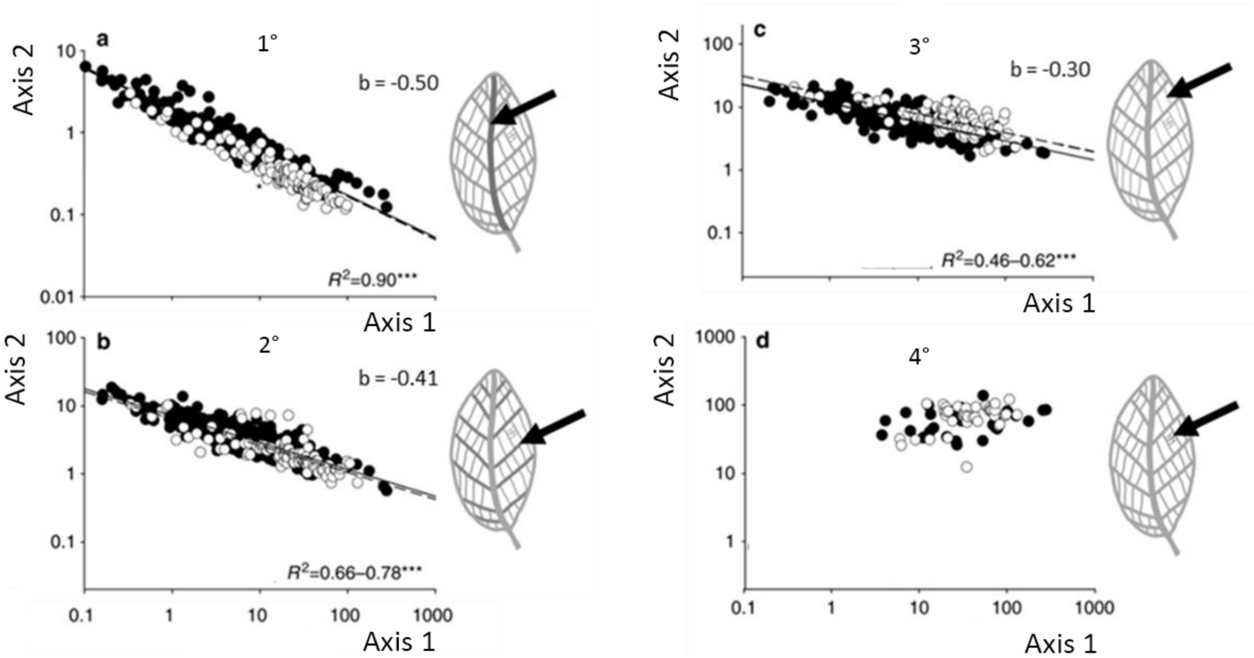


Fig.1. Axis 1 = Leaf area (cm^2). Axis 2 = Vein density (= vein length per leaf area; $cm \cdot cm^{-2}$).

Based on these data, a model was created (Figure 2) which is generally true for most of the species examined. It details the change of various structural features during the development of a dicotyledonous leaf.

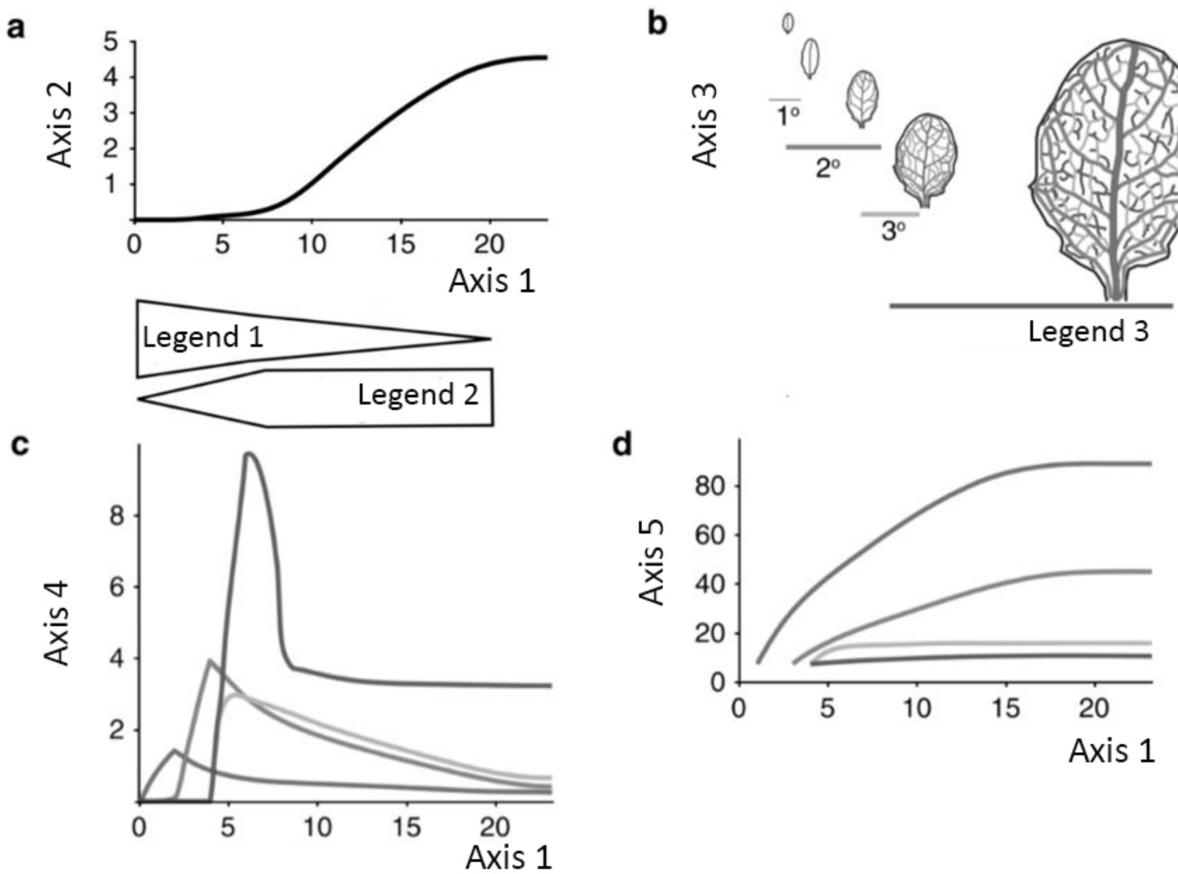


Fig.2. Axis 1 = Days of leaf expansion. Axis 2 = Leaf area (cm^2). Axis 3 = Vein development. Axis 4 = Vein density ($mm \cdot mm^{-2}$). Axis 5 = Vein diameter (mm). Legend 1 = Cell division growth. Legend 2 = Cell expansion growth. Legend 3 = Minor veins.

Determine the relationship between the following pairs (A and B). Indicate your answer on the answer sheet by using the symbols > or < or =.

Q.22.1 A. Vein density scaling with leaf surface area for midvein and 2°.
B. Vein density scaling with leaf surface area for 3° and 4°.

Q.22.2 A. Vein density scaling with leaf surface area in species with branched veins.
B. Vein density scaling with leaf surface area in species with parallel veins

Q.22.3 Identify how density of veins of different hierarchical levels change over time and label Fig.2.c below accordingly. Match the letters below (A-D) with the appropriate roman numeral on your answer sheet. Indicate your answer by putting an X in the appropriate box on the answer sheet.

A. 1° veins
B. 2° veins
C. 3°veins
D. Minor veins

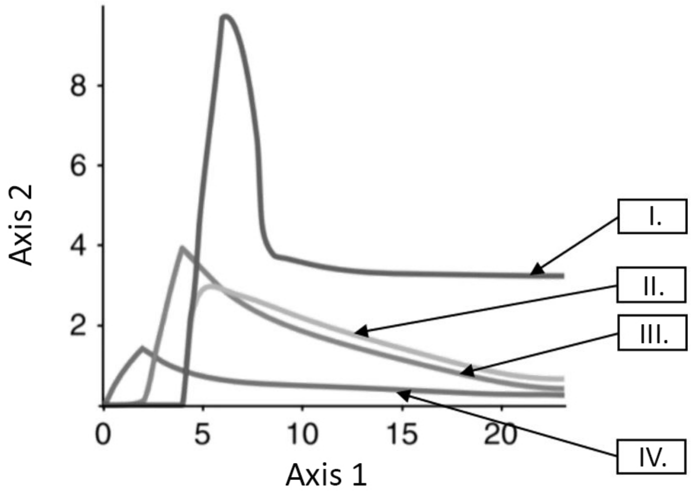


Fig.3. Axis 1 = Days of leaf expansion. Axis 2 = Vein density ($mm \cdot mm^{-2}$)

Genetics and Evolution

Q23

The gene that determines red or black fur colour is X chromosome linked. During female embryonic development one of the two parental copies of chromosome X becomes inactivated. If a female cat inherits different alleles from its parents, we can observe the mosaicism. For example, if the maternally derived X chromosome becomes inactivated and the paternally remains active such cells form black patches on the cat's coat. If the paternally derived X chromosome becomes inactive and the maternally derived remains active such cells form red patches. (The white patches originate through a different mechanism.)

Fig.1 - Kittens

You have a list of possible observations regarding the inheritance of red/black coat colour:

A. There are no smaller red spots in the middle of the large black spots or vice versa.

B. Black spots are predominantly located towards the back of the animals

C. The mosaic spots are large.

D. The red/black spot boundaries are sharp.

E. The feet of the cats are usually white.

F. The distribution of spots in different cats is very variable.

Match the conclusions (Q23.1-4) with one of the observations listed above. Only one letter should be matched with one conclusion and one letter may only be used once. Indicate your answer by putting an **X** in the appropriate box on the answer sheet.

Q.23.1 X chromosome inactivation takes place during early embryogenesis.

Q.23.2 Inactivation of the maternal or paternal X chromosome is a random event.

Q.23.3 X chromosome inactivation is irreversible.

Q.23.4 X chromosome inactivation is a cell autonomous process, i.e. the neighboring cells do not influence it.

Q24

Carolyn does not know who the father is of her daughter Lucy. She knows that the father is one of the five members of the Happening rock band. You collected DNA samples from the seven people listed in the table and analyzed five (A through E) of the VNTR sequences. (The VNTRs - variable number tandem repeats - are DNA sequences present in variable number of tandemly arranged copies in both autosomes and sex-chromosomes) The figures in the table represent the number of copies in the different VNTRs.

VNTR	Carolyn	Lucy	Frank	Conrad	Andrew	Bill	Lewis
A	7 + 8	2 + 7	3 + 3	5 + 7	2 + 8	3 + 7	2 + 3
B	4 + 5	4 + 6	4	7	6	4	6
C	11 + 18	9 + 11	9 + 10	8 + 13	10 + 11	9 + 12	9 + 13
D	7 + 19	7 + 21	7 + 21	9 + 15	7 + 21	8 + 15	15 + 21
E	6 + 10	6 + 12	6 + 9	9 + 12	17 + 17	6 + 12	12 + 17

Table 1

Q.24.1 Indicate Lucy's father with an **X.**
A. Frank
B. Conard
C. Andrew
D. Bill
E. Lewis

Q.24.2 Carolyn and Frank have a son called Rupert. List all possible VNTRs copy numbers Rupert can have at the analysed VNTR loci (A-E). Treat the rows independently from one another. If an individual has two alleles of a specific VNTR locus, the number of repeats in each allele should be indicated with a '+' sign between them.

Q25

The *Eco*RI restriction enzyme recognizes the 5’G*AATTC3’ nucleotide sequence and cuts the DNA at the “*” site. You have a plasmid preparation with billions of copies of a plasmid composed of 1.3×10^4 base pairs. Let's assume that the sequence of base pairs is random in the plasmid DNA.

Q.25.1 On average, how many DNA fragments form following the digestion of the plasmid DNA by the *Eco*RI restriction enzyme? Give your answer rounded to the nearest integer.

Q26

A single base pair deletion mutation occurred in a short gene, which changes the MATE peptide's amino acid sequence to MATEK. The peptide sequence is given based on the one letter codon table. You can find the original peptide and nucleotide sequence below, but some nucleotides are missing.

PS	M			A			T			E			(STOP)					
NS	A	T	G	G	C	T	A	C	T	G	A	A	T	*1*	*2*	*3*	*4*	G	*5*

Fig.1. Original sequence. PS = Peptide sequence. NS = Nucleotide sequence.

Q.26.1 Give the letter of the missing nucleotides (1-5). If in any one position more than one type of nucleotide can occur, give only one of them.

Q27

Karolyn and Fred visit your genetic counselling office. They tell you the data summarized in the table. You aware of the fact that:

(i) the absence of patella is caused by the dominant mutation Np (Nail-patella).

(ii) The Np mutant allele (and of course the Np^+ wild-type gene) and the ABO blood group gene (with alleles I^A, I^B and i) are inherited on chromosome 9 and are 18 cM apart.

Person	Does she/he have patella?	Blood group
Karolyn	No	B
Karolyn's mother	Yes	B
Karolyn's father	No	0
Fred	Yes	0
Fred's mother	Yes	A
Fred's father	Yes	0

Determine Karolyn's and Fred's genotypes. The alleles on one side of '/' sign are linked. (For example, in case of $Np\,i$ / $Np^+\,I^A$ genotype, Np and i alleles are linked and Np^+ and I^A alleles are linked.)

Q.27.1 What is Karolyn's genotype? Indicate your answer with an **X**.
A. $Np\,I^B$ / $Np\,I^B$
B. $Np\,I^B$ / $Np^+\,i$
C. $Np\,I^B$ / $Np\,i$
D. $Np^+\,I^B$ / $Np\,i$
E. $Np^+\,i$ / $Np\,I^B$

Q.27.2 What is Fred's genotype? Indicate your answer with an **X**.
A. $Np\,i$ / $Np\,i$
B. $Np^+\,i$ / $Np^+\,i$
C. $Np^+\,i$ / $Np^+\,I^A$
D. $Np\,i$ / $Np^+\,I^A$
E. $Np^+\,I^A$ / $Np^+\,I^A$

Q.27.3 Karolyn is pregnant and Fred is the father. Her foetus has blood group B. Add the probability in percentage that her child will have no patella.

Q28

The a, the b and the c recessive mutations (with the corresponding a^+, b^+ and c^+ wild-type alleles) are linked in an autosome, i.e. they belong to the same linkage group. From a genetic cross in which females heterozygous for the *a*, the *b* and the *c* recessive mutations were mated with males that were homozygous for *a*, *b* and *c* the following phenotypic offspring emerged (with the number of offspring):

ab^+c – 40;

abc^+ – 12;

$a^+b^+c^+$ – 3;

a^+b^+c – 13;

ab^+c^+ – 20;

a^+bc^+ – 41;

a^+bc – 19;

abc – 2.

Q.28.1 What is the sequence of the genes? Choose the letter of the correct sequence!
A. a-b-c
B. a-c-b
C. b-a-c

Q.28.2 Indicate double recombinant offspring phenotypes with an **X**, and all other phenotypes with **O.**

Q.28.3 Determine in cM units the distance between the given gene pair. Round your answer to the nearest integer.
b - c

Q29

A solution of linear DNA with billions of copies of the same sequence was distributed into eight equal samples. The samples were digested with the restriction enzymes as indicated in the top of the figure. All samples were loaded on a gel for size determination by gel electrophoresis. The figure shows the results of gel electrophoresis. The numbers give the number of base pairs (in thousands) of the fragments. Please construct restriction map (also called physical) of the studied DNA by placing the so-called restriction sites of the three enzymes on the line on the answer sheet The restriction map shows potential positions (boxes above the line) of the restriction sites (i.e. the positions where the restriction enzymes cut the DNA). Indicate which box represents which restriction site. Not all box represents actual restriction sites! Please notice that one of the bands in the "All three" lane is twice as thick as any of the others. All digestion was complete.

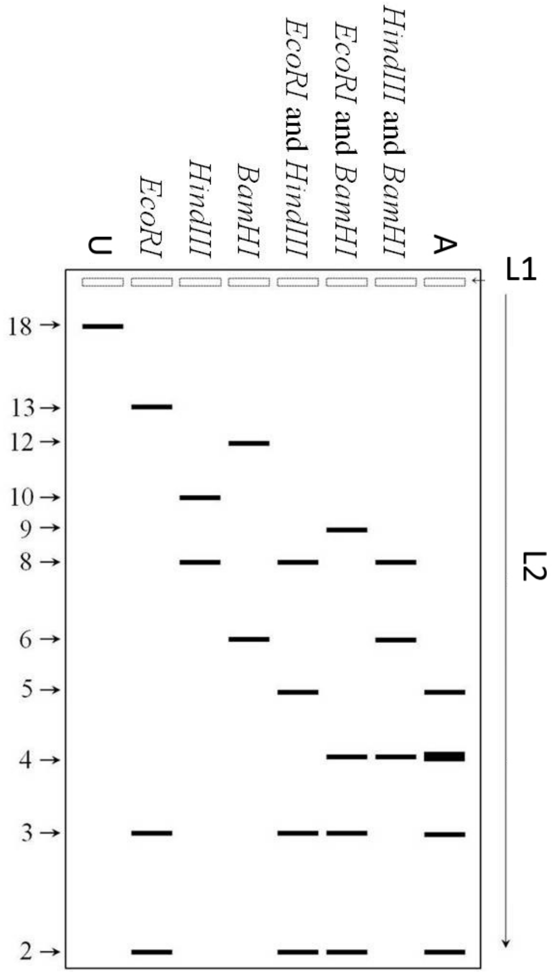


Fig.1. U = Undigested. A = All three. L1 = Pockets. L2 = Direction of gel electrophoresis

Q.29 Indicate which box represent which restriction site. Use the following notation:
E – EcoRI restriction site
H – HindIII restriction site
B – BamHI restriction site
X – no restriction site here

Q30

DNA constantly accumulates mutations both during development and adult life, therefore several pathways exist which are meant to repair these damages. People with mutations in an autosomal gene encoding a DNA repair enzyme acquire more mutations in all chromosomes, including the X and Y chromosomes. However, the Y chromosome has fewer essential genes compared with the X chromosome. Three hypothetical couples are presented on your answer sheet.

Dr = homozygous mutant individuals, + = wild-type, M = male, F = female.

Q.30	Indicate the expected relative number of female (F) versus male (M) offspring these couples may have surviving into adulthood. Use the following symbols in the cells provided: =, <, >.

Ecology

Q31

The introduced black pines have displaced some dolomite rock grasslands in Hungary. Due to the shadowing effect of the pine trees and to the substances leached out of the litter layer, these pine forests almost completely changed the originally rich flora of rock grasses. In the 0th year of the experiment, forest fires devastated black pines in the Buda Mountains. Researchers wanted to find out whether the original rock grass community would be restored.

Number of species (Figure 1A) and % coverage over time (Figure 1B) was measured. Sites in the burnt area on the North (N) and South (S) slopes of the mountains and also natural rocky grasslands as controls (NC & SC) were compared. They monitored the areas (except control) for 10 years.

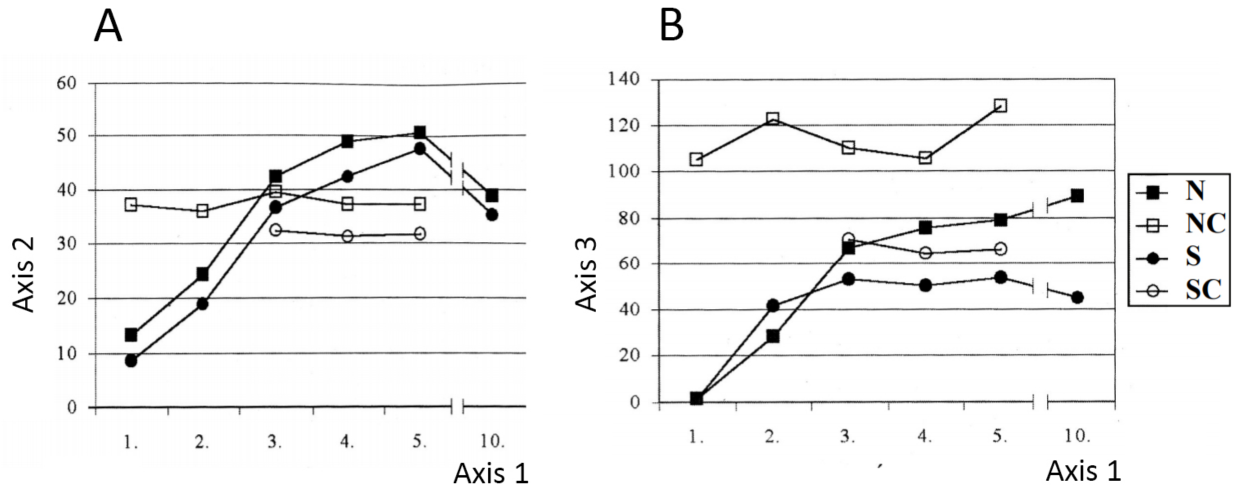


Fig.1. A = Average number of species found in the sampling squares. B = Average cumulative percentage of the area covered by plants. Axis 1 = Year. Axis 2 = Number of species. Axis 3 = Percentage

During the process researchers could detect different changes in the composition of plant community, which is shown in Figure 2.

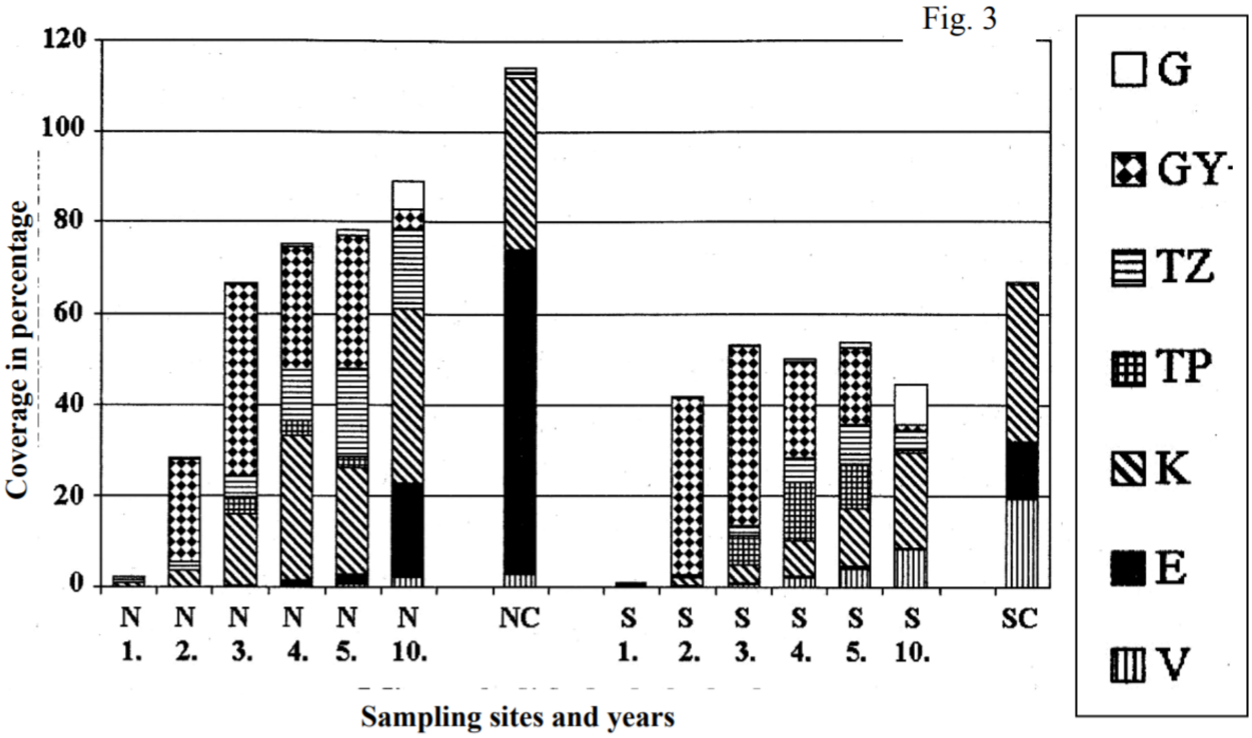


Fig.2. Composition of Plant Community. Axis 1 = Sampling sites and years. NC = northern control area. SC = southern control area. G = Economical plants (mainly black pine here). GY = Weed plants (heavy disturbance-tolerant, often invasive species); TZ = Natural (weak) disturbance-tolerant species. TP = Pioneer species (first colonialists on the parent rock). K = Accompanying species. E = Dominant species of the community. V = Protected species

What can be the cause of the species richness drop between year 5 and 10? Indicate with an **X** if each the following statements are true (T) or false (F).

Q.31.1 The invasive weeds suppressed the native species.

Q.31.2 The competition between the plants grew stronger between year 5 and 10 so the weak competitor species have disappeared.

Q.31.3 The soil ran out of nutrients which caused a plant biomass decrease which caused the decline of the species number.

Q.31.4 The main cause can be the almost complete disappearance of weeds.

Q.31.5 The cause of the decline in species number is a stronger competition between plants than in natural grasslands.

Q32

Population dynamics models can be used to predict changes in population size. The Beverton-Holt model assumes discrete (not overlapping) generations. The model is given below:

$$N_{t+1} = \frac{BN_t}{1 + \alpha N_t^m}$$

N_t: The population size (number of individuals) in generation t.

N_{t+1}: The population size (number of individuals) in the next generation (generation t+1).

B: The maximal growth of population size per individual. α, m: Parameters that determine the effect of population size on per-capita rate of increase.

One of the advantages of the Beverton-Holt model is that it can describe both exponential and logistic growth (growth which slows to a plateau) depending on the value of the parameters.

Q.32.1 You can change the model to produce exponential growth by setting only one of the parameters B, α or m to a fixed value. There may be more than one option for this. State the parameters which must be fixed and the value or values they may take. Find all the possible parameter sets. Write the letter of the parameter(s) to column A and the value(s) to column B in the table in the Answer Sheet. Mark any potential unused cells with an **X**.

Q.32.2 The equilibrium population (N_{eq}) is a population size which remains constant with time. How can the equilibrium population size be calculated using the Berverton-Holt model? Indicate your answer with an **X.**

A. $N_{eq} = \sqrt[m]{\frac{B-1}{\alpha}}$

B. $N_{eq} = \frac{\frac{\sqrt[m]{N_t}}{\alpha} - 1}{B}$

C. $N_{eq} = \sqrt[m]{\frac{\alpha+1}{B}}$

D. $N_{eq} = \sqrt{\frac{B-1}{\alpha}}$

E. $N_{eq} = (\frac{\alpha+1}{B})^m$

Q33

A dolina is a type of sinkhole common in Hungary home to important species. The goal of an experiment was to determine how the size of the dolina affects species richness in the Mecsek Mountains (Figure 1) and Aggtelek Karst (Figure 2). A = all species. B = Beech forest species (like average temperature conditions with plenty of rain). C = oak forest species (like warmer and drier conditions). D = cool environment specialists.

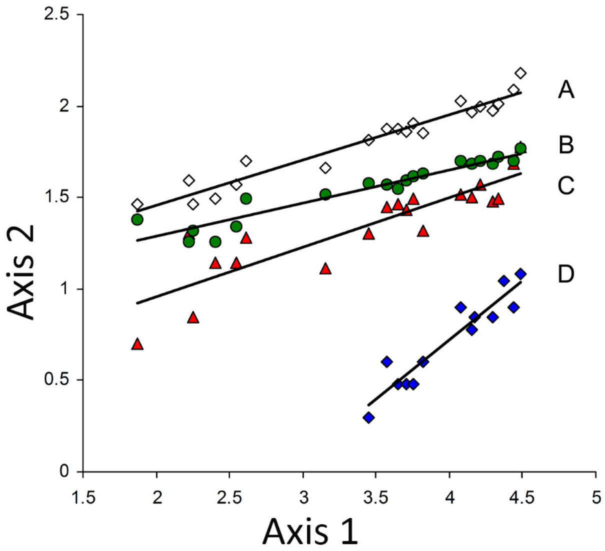


Fig.1. The correlation between the area of the dolina and the species richness in the Mecsek Mountain. The values of the axes are in $\log_{10}$ transformed. Axis 1 = log area (m^2). Axis 2 = log species richness

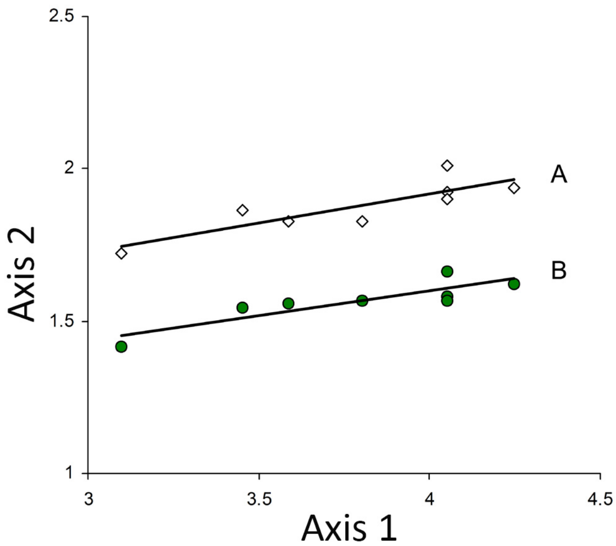


Fig.2. The correlation between the area of the dolina and the species richness in the Aggtelek Karst. The values of the axes are in log10 transformed. Here the oak forest species and cool-adapted species are not shown separately, because there was no significant correlation between their species richness and the size of the dolina. Axis 1 = log area (m^2). Axis 2 = log species richness

Indicate with an **X** if each the following statements are true (T) or false (F).

Q.33.1	In the Mecsek Mountains the number of cool-adapted species grows faster with increasing dolina size, than the number of all species.
Q.33.2	According to the data below a certain dolina size, no cool-adapted plant was found.
Q.33.3	The proportion of beech forest species relative to all species increases with increasing dolina size in the Aggtelek Karst.
Q.33.4	The number of beech forest species compared to all is affected by dolina size but not by geographic location
Q.33.5	Regardless of size, every dolina plays an equally important role in conservation of rare plant species.
Q.33.6	Most of the examined Mecsek Mountains dolinas are in oak forest covered areas.
Q.33.7	According to the data, the bigger dolinas have positive effect only to the cool-adapted plants.

Q34

Scientists studied how abandoned agricultural fields become mature grassland through succession. They studied lands which were partly colonised by species from surrounding natural grasslands. They classified the plots of lands into those colonised mainly by grasses which dominate natural grassland (D), those colonized mainly by grasses which are subordinate in natural grassland (S1), those colonized mainly by subordinate dicots in natural grassland (S2), and those colonized mainly by alien weeds (A).

For different types of land they measured the following values:

(a) Diversity of species

(b) Equitability (evenness) index: The lower the value, the bigger the difference between the abundance of the different species. An index of 1 indicates equal numbers of individuals from all species.

(c) Similarity (Bray-Curtis): Measures how similar the abandoned land is to natural grassland based on the abundance of the species.

$$BC = \frac{2L}{M+N}$$

BC: Bray-Curtis similarity

L: Is the sum of only the lesser abundance (in this case coverage) of the species found in both place.

M: Total abundance (coverage) of species found in the first community.

N: Total abundance (coverage) of species found in the second community.

(d) Similarity (Sorensen): Measures how similar the abandoned land is to natural grassland based on the presence of the species.

$$Ss = \frac{2a}{2a+b+c}$$

SS: Sorensen similarity

a: the number of species found in each community

b: the number of species found just in the first community

c: the number of species found just in the second community

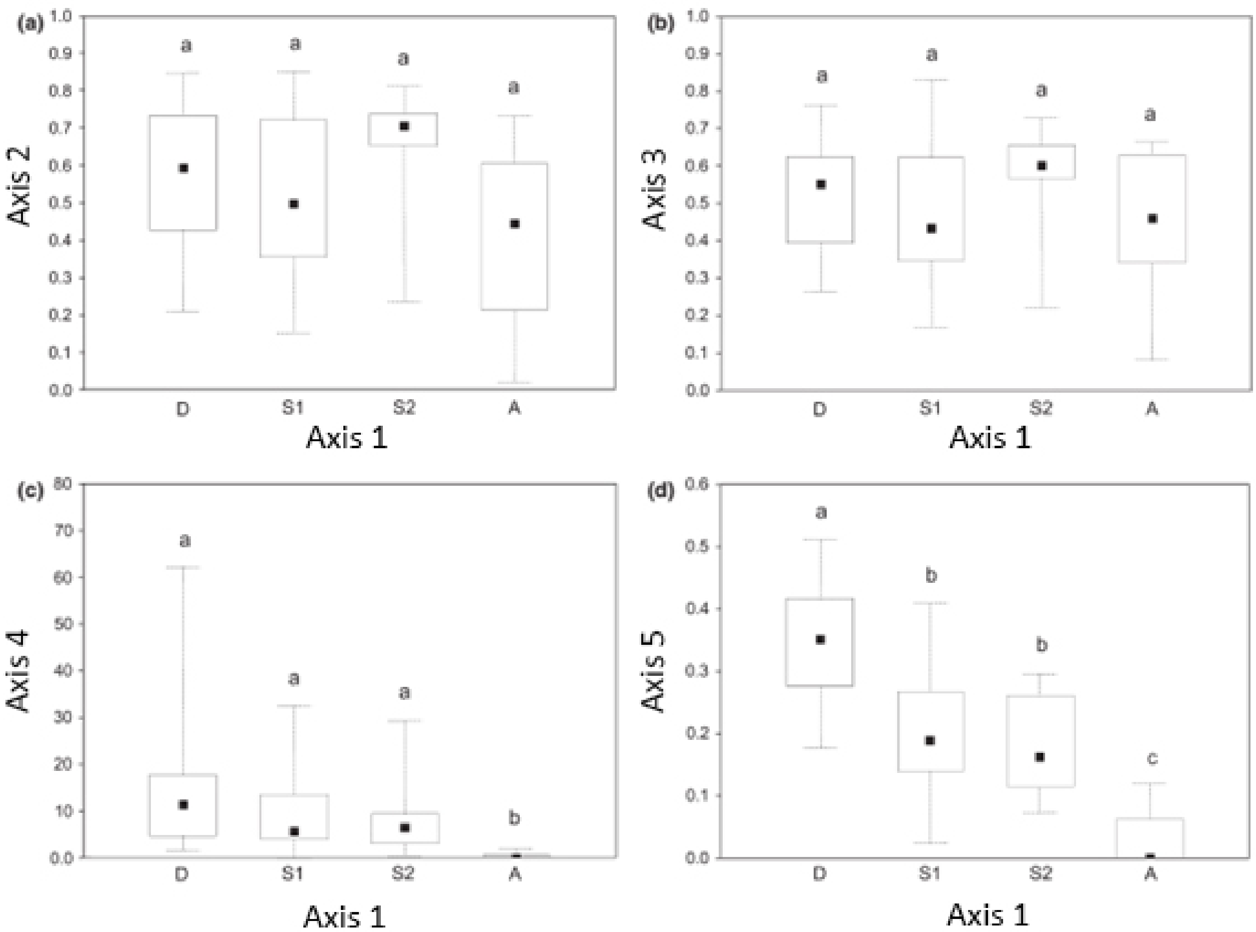


Fig.1. The measured values for different land types. (Values for a, b, d between 0-1, for c between 0-100%.) Letters above bars indicate statistical significance, where two bars with same letter are not statistically different, but different letters are. Axis 1 = Groups. Axis 2 = Species diversity. Axis 3 = Similarity (Bray-Curtis). Axis 4 = Equitability. Axis 5 = Similarity (Sorensen). D = Species that are dominant in natural grasslands. S1 = Subordinate grasses in natural grasslands. S2 = subordinate dicots in natural grasslands. A = alien weeds

Indicate with an **X** if each the following statements are true (T) or false (F).

Q.34.1 In grasslands the species D as a dominating species supresses the others more than S1 and S2 do.

Q.34.2 In this study grasslands dominated by alien weeds (A) had lower species diversity.

Q.34.3 Results show, that in alien weed (A) dominated communities only few species have high abundance, and all the other has very low compared to the other communities.

Q.34.4 Based on abundance of species, the D species dominated communities are definitely more similar to surrounding natural grasslands, than S1 and S2 communities.

Q.34.5	The (c) and (d) graphs show that among these communities the S1 and S2 types are the most similar.

Q.34.6	There are communities with no common species with the surrounding natural grasslands.

Q.34.7	There are some S2 communities, which have only some rare common species with the surrounding natural grasslands.

Biosystematics

Q35

The following question is about nutritional strategies and levels of organization in Eukaryotes.

Q.35 Write the numbers of the species or group of living organisms listed below (1-8) into the appropriate spaces of the figure on your answer sheet.

1. Amoeba (*Amoeba proteu*s)
2. *Chlamydomonas* (single-celled green algae) in presence of light
3. Baker's yeast (*Saccharomyces* sp.)
4. Nitrifying bacteria, which rely on the energy deriving from oxidising nitrogen containing inorganic compounds for fixing/reducing atmospheric carbon dioxide.
5. Fresh water sponge
6. Bacterium causing pneumonia
7. Sporophyte of ferns (e.g. *Pteridium* sp.)
8. Protonema of mosses (e.g. *Sphagnum* sp.)

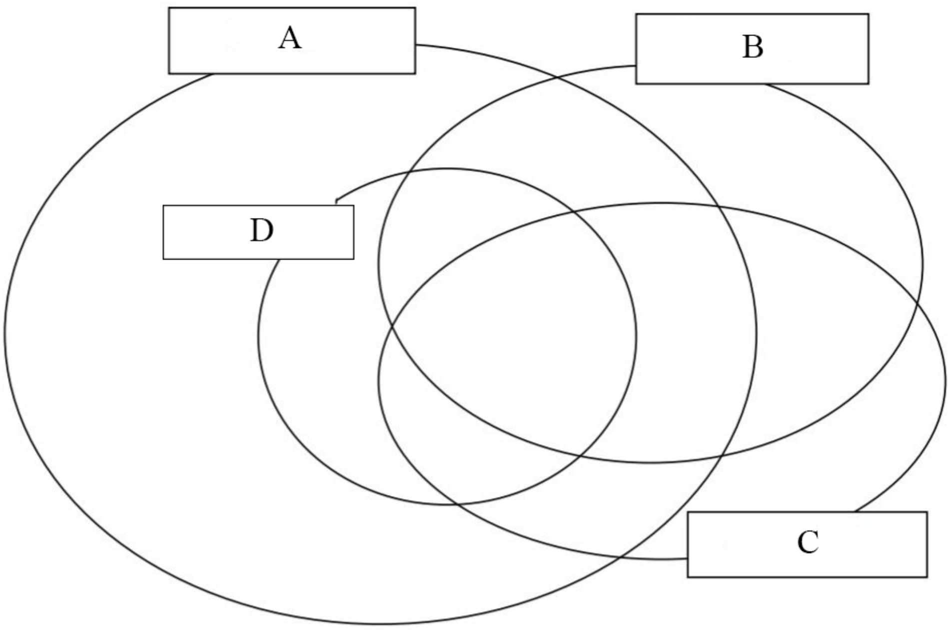


Fig.1. A = Eukaryotes. B = Heterotrophs. C = Chemotroph. D = Multicellular, but below tissue level

Q36

A Hungarian research group examined the diversity of biochemical processes of a group of fungi called dark septate endophytes (DSE). They used api-ZYM, a biochemical assay for detecting the activity of enzymes, the results of which is illustrated in Figure 1. They want to use these to create an identification key based on the enzyme activities of the species.

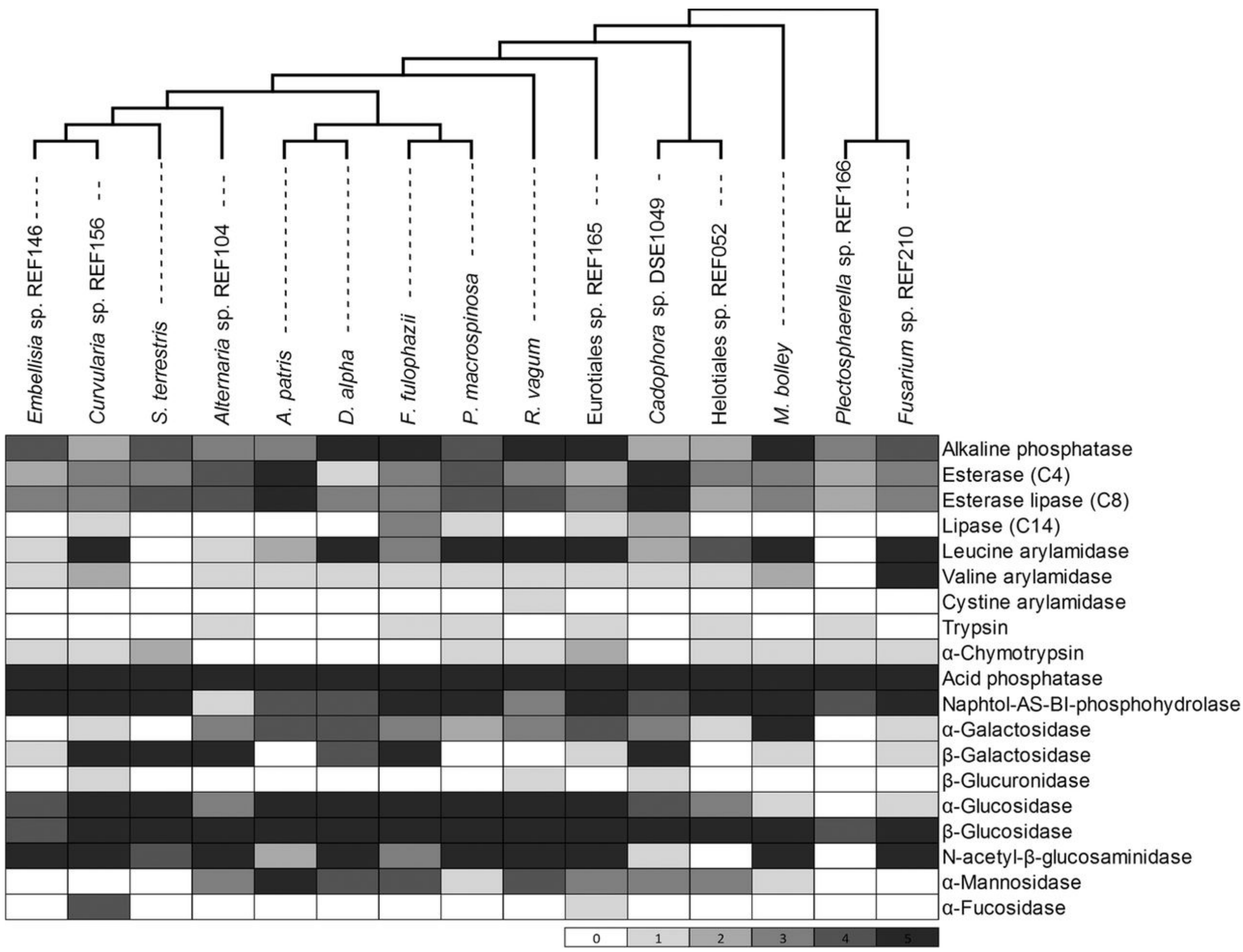


Fig. 1. The phylogenetic tree and the measured enzyme activities of the examined fungus species. The darker color shows stronger enzyme activity. (As shown below the table: 0 – no enzyme activity, 5 – maximal enzyme activity.)

Q.36.1 Which enzyme's activity gives us no information about identifying the examined species? Indicate your answer by putting an **X** in the appropriate box on the answer sheet.
A. Esterase (C4)
B. Valine arylamidase
C. Cysteine arylamidase
D. Acid phosphatase
E. β-glucosidase

We created an identification key for some of the examined species, but the names are missing. The roman numerals indicate the species in the key.

1. a) The leucine arylamidase activity is very high (5). ***Go to 2.***

b) The leucine arylamidase activity is lower (0-4). ***Go to 5.***

2. a) The β-galactosidase activity is very high (5). ***I.***

   b) The β-galactosidase activity is lower (0-4). ***Go to 3.***

3. a) The esterase lipase (C8) activity is high (4 or 5). ***II.***

   b) The esterase lipase (C8) activity is lower (0-3). ***Go to 4.***

4. a) The α-galactosidase activity is 4. ***III.***

   b) The α-galactosidase activity is 1. ***IV.***

5. a) The β-galactosidase activity is very high (5). ***Go to 6.***

   b) The β-galactosidase activity is lower (0-4). ***Go to 7.***

6. a) The α-glucosidase activity is 3. ***V.***

   b) The α-glucosidase activity is 4. ***VI.***

7. a) The N-acetyl-β-glucosaminidase activity is 5. ***VII.***

   b) The N-acetyl-β-glucosaminidase activity is 2. ***VIII.***

Q.36.2 Match the roman numerals in the key (I-VIII) with the letters of the species below (A-L). Only one species (letter) belongs to each numeral. Indicate your answer by putting an **X** in the appropriate box on the answer sheet.

A – *D. alpha*

B – *Fusarium* sp. REF210

C – *Helotiales* sp. REF052

D – *S. terrestris*

E – *Altermaria* sp. REF104

F – *Curvularia* sp. REF156

G – *A. patris*

H – *Embellisia* sp. REF 146

I – *M. bolley*

K – *Cadophora* sp. DSE1049

L – *R. vagum*

Ethology

Q37

Cuckoos lay their eggs in the nests of Hungarian songbirds which unwittingly raise the cuckoo hatchling. Host parents will destroy a cuckoo egg if they notice it, but cuckoo eggs look similar to host eggs.

Scientists painted eggs of Reed Warblers (a common songbird in Hungary) to look like cuckoo eggs. The different treatments are described in Fig 1.

Scientists counted how many eggs the Reed Warblers destroyed. (Fig.2) In control nests, there were no rejections. In the “parasitized” nests, the warblers rejected some of the eggs. Most of the rejected eggs were painted, but sometimes the birds rejected their unpainted eggs also.

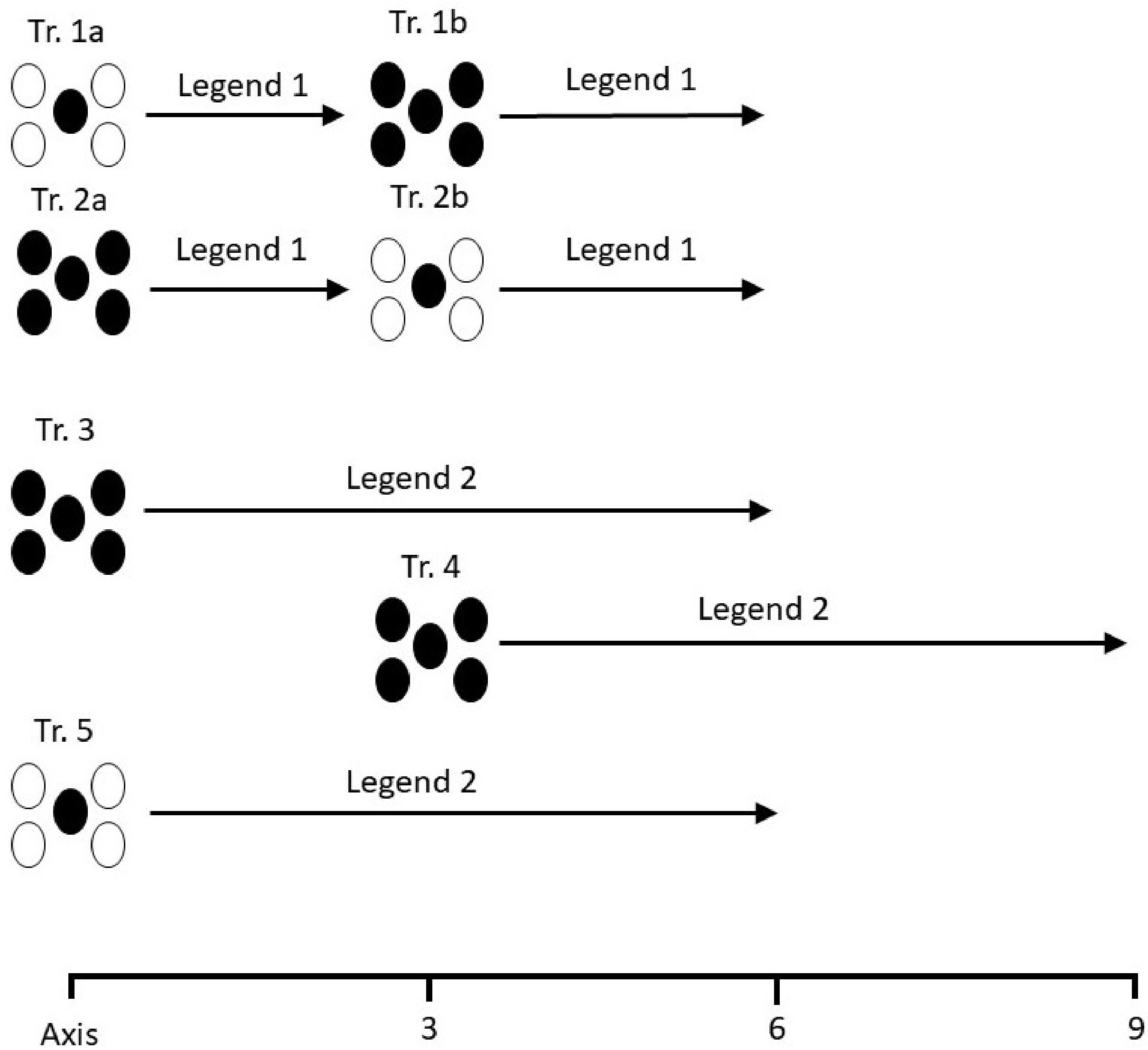


Fig.1. The schematic representation of treatments (Tr.) applied. White eggs denote unpainted (‘host’) eggs, and dark eggs denote painted (‘parasitic’) eggs. Monitoring periods after parasitism (3 or 6 days) are also shown. Axis = day of clutch completion. Legend 1 = 3d monitoring. Legend 2 = 6d monitoring.

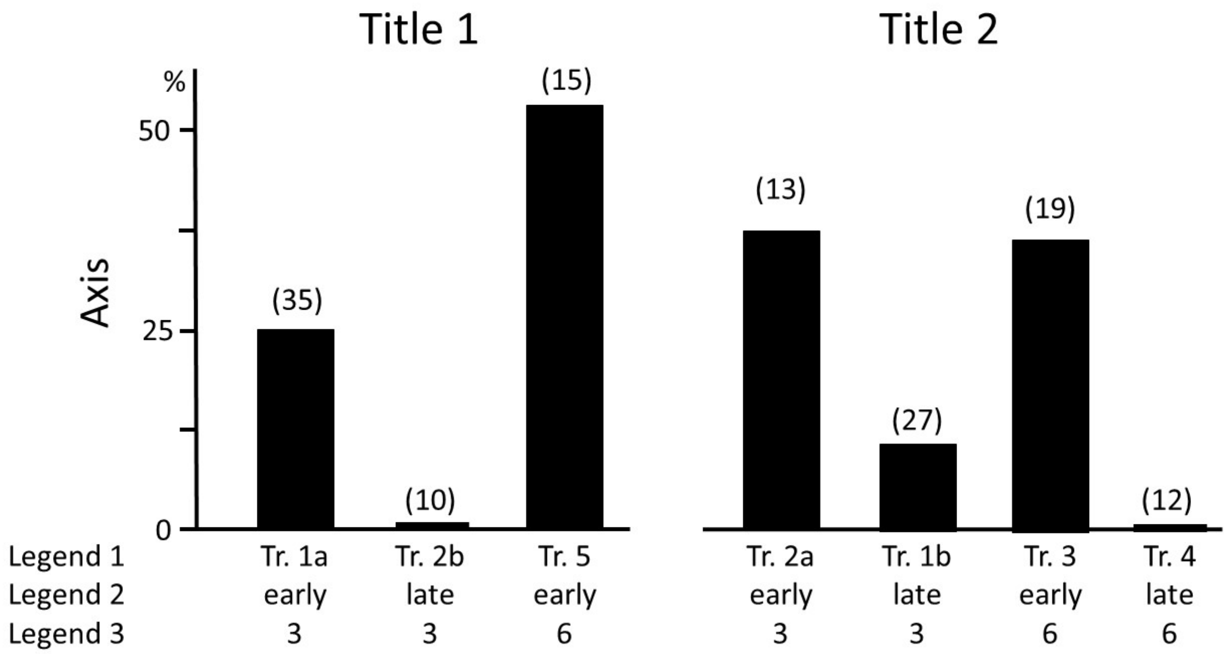


Fig. 2. The rejection rate of the eggs (y axis, %) in each treatment (x axis). Title 1 = Single parasitism. Title 2 = Multiple parasitism. Axis = Rejection rate. Legend 1 = Treatment. (The numbering of the treatments is the same as in Fig. 1.) Legend 2 = Start of experiment: early = immediately after clutch completion, late = 3 days after clutch completion. Legend 3 = Monitoring period (days).

Based on the information above, indicate with an **X** if each of the following statements is true (T) or false (F).

Q.37.1 The rejection rate is higher during the earlier stages (5-8 days), than later (after 8th day).

Q.37.2 Adding one parasite egg early increased the rejection rate for later parasite eggs.

Q.37.3 The probability of the host ejecting its own egg is higher in the later period, than during the egg laying period.

Q.37.4 The later the cuckoos lay their eggs, the higher chance their offspring have to be rejected.

Q38

It can be difficult to determine the strength of natural selection on traits in wild fish. Scientists measured the length of Atlantic silverside (*Menidia menidia*) before winter (dark bars) and after winter (grey bars), in three different locations. Assume that changes in length can be attributed to differential mortality.

Col 1	Col 2	Col 3		Col 4
		Col 3A	Col 3B	
Nova Scotia (NS)	44°40’ N	29 Sep 1987 26 Sep 1988	30 May 1988 3 Jun 1989	244 250
New York (NY)	42°45’ N	27 Oct 1987 25 Oct 1988	27 Apr 1988 18 Apr 1989	183 175
South Carolina (SC)	33°20’ N	1 Dec 1987 28 Nov 1988	30 Mar 1988 20 Mar 1989	120 112

Table 1. Sample sites of the experiment. Col 1 = Site (State). Col 2 = Latitude. Col 3 = Sample dates. Col 3A = Autumn. Col 3B = Spring. Col 4 = Winter duration (days)

The results of the two years and three sample sites are shown in Fig. 1.

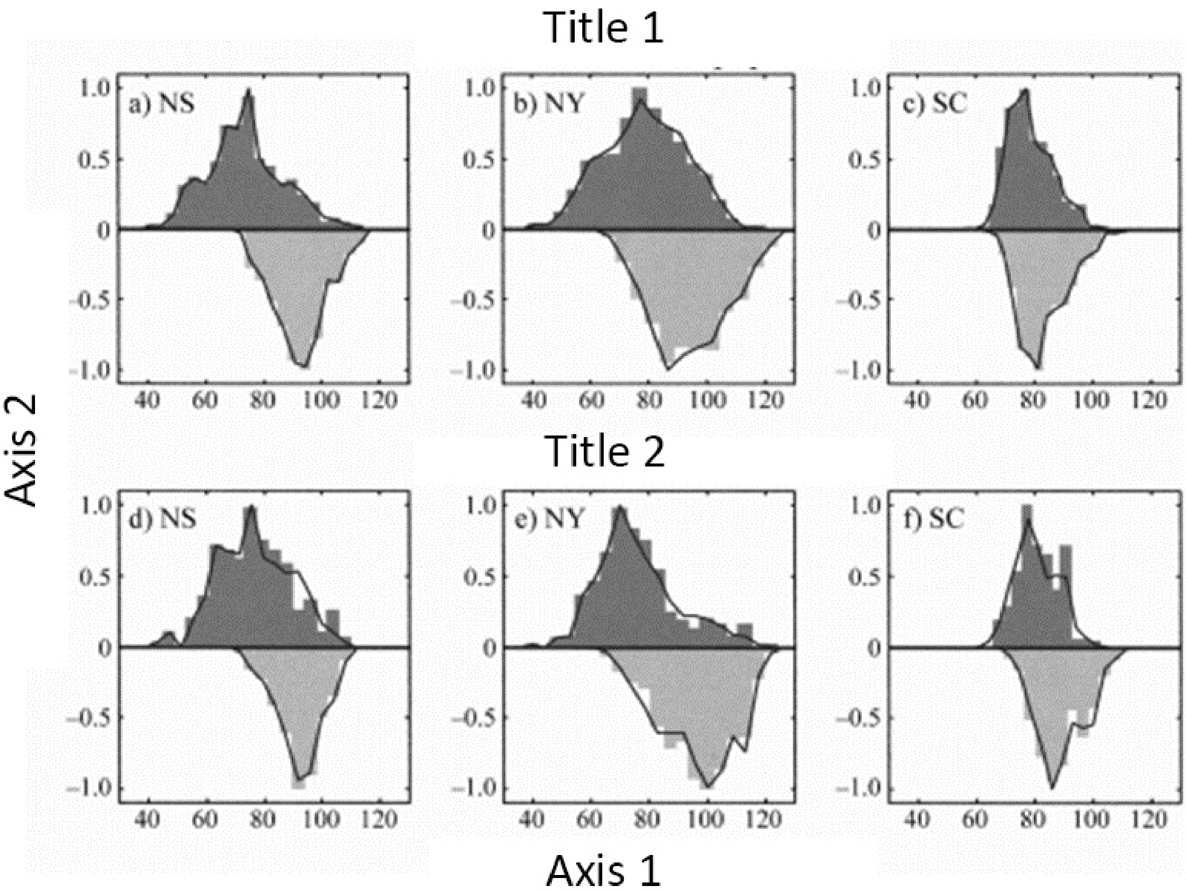


Fig. 1. The size distribution of *Menidia menidia* in the three sample sites (the meaning of abbreviation is shown in Table. 1.) The dark, upward bars show the autumn results, and the light downward bars show the spring results. Title 1 = 1987-1988 *Menidia* populations. Title 2 = 1988-1989 *Menidia* populations. Axis 1 = Total length of *Menidia* (mm). Axis 2 = Relative frequency of *Menidia*.

Based on the information above, indicate with an **X** if each of the following statements is true (T) or false (F).

Q.38.1 According to the results, the 1987-88 winter should be colder in New York (NY) than the 1988-89 winter.

Q.38.2 The longer and colder winter results stronger selection against smaller specimens.

Q.38.3 Size may affect the success of migration, which can help survival.

Q.38.4 The smaller sized fishes of the species have bigger chance to survive unfavorable conditions.

Fig. 2 shows the relative rate of survival at two sample sites in two different years.

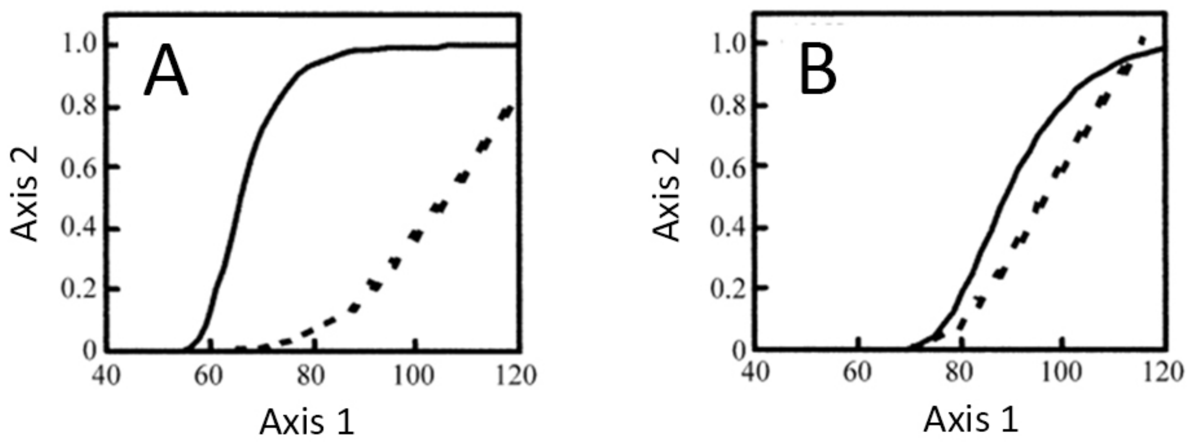


Fig. 2. The relative survival of *Menidia* as function of the total length (body size) from two sample sites. The solid lines indicate the 1987-88 year, while the dashed lines the 1988-89 year. Axis 1 = Total length of *Menidia* (mm). Axis 2 = Relative survival of *Menidia*

Q.38.5 Match the curves in Figure 2 (A and B) with the sample sites. Indicate your answer by putting an **X** in the appropriate box on the answer sheet.

- Nova Scotia = NS
- New York = NY
- South Carolina = SC

END OF THE THEORY 2